EXP 801 DS Wheat Heat Drought

— Ed Chaplin (USyd), *Apr–June 2025* —

Analysis of anatomical traits

Load packages

```
library(knitr)
#knitr::spin("Exp801.anat.anout.v2.r")
library(ggplot2)
library(asreml)
library(asremlPlus)
```

Set up constants and files

```
options(width=95)
load("../data/anat.Analyses.v2.RData")
source("Exp801.anat.anal.v1.stub.r")
```

Show the results

```
for (resp in responses.anat[kvec]) {
  showIndvModelResults(models[[resp]], response=resp)
}
```

```
#### Results for StomatalCount


  Final model converged



#### Formulae from asreml object

fixed: StomatalCount ~ ASide/ABlock + Rm_Temperature/(AWatering * ALine +
     HWatering * HLine)
random: NULL
residual: ~ (dsum(~ units | Rm_Temperature))




####  Summary of the fitted variance parameters for StomatalCount

                           component std.error  z.ratio bound %ch
Rm_Temperature_7,Heat!R     10.44401  1.959970 5.328658     P   0
Rm_Temperature_8,Ambient!R  22.31792  4.114156 5.424665     P   0


####  Pseudo-anova table for fixed terms
```

Wald tests for fixed effects.
Response: StomatalCount

	Df	denDF	F.inc	F.con	Margin	Pr
(Intercept)	1	101.6	7756.0	7756.0		0.0000
ASide	1	101.6	2.2	2.0	B	0.1586
ASide:ABlock	2	101.7	5.7	6.0	C	0.0035
Rm_Temperature	1	104.0	0.1	0.1	A	0.7795
Rm_Temperature:AWatering	1	58.9	0.9	0.9	B	0.3359
Rm_Temperature:ALine	9	58.9	0.9	0.9	B	0.5552
Rm_Temperature:AWatering:ALine	9	58.9	0.8	0.8	C	0.5940
Rm_Temperature:HWatering	1	56.8	0.9	1.1	B	0.3067
Rm_Temperature:HLine	9	56.8	7.1	7.1	B	0.0000
Rm_Temperature:HWatering:HLine	9	56.8	1.7	1.7	C	0.1123

Sequence of model investigations for StomatalCount

(If a row has NA for p but not denDF, DF and denDF relate to fixed and variance parameter numbers)

	terms	DF	denDF	p	AIC	BIC	action
1	Initial model	43	1.0	NA	649.460213	784.491998	Starting model
2	Start with residual heterogeneity in Watering & Rm_Temperature for Residual	NA	NA	NA	NA	NA	Changed residual
3	Try homogeneity in Rm_Temperature for Residual	0	-2.0	NA	4.283150	-1.224031	Unswapped
4	Try homogeneity in Watering for Residual	0	-2.0	NA	-3.861354	-9.368534	Swapped
5	Room:Side	1	NA	NA	NA	NA	Boundary
6	Room:Side:Location	1	NA	NA	NA	NA	Boundary
7	Room:Side:Location:SubUnit	1	NA	NA	NA	NA	Boundary
8	Fix bound variance components	0	0.0	NA	0.000000	0.000000	Swapped
9	Rm_Temperature:HWatering:HLine	9	56.8	0.1123	NA	NA	Nonsignificant
10	Rm_Temperature:HLine	9	56.8	0.0000	NA	NA	Significant
11	Rm_Temperature:HWatering	1	56.8	0.3067	NA	NA	Nonsignificant
12	Rm_Temperature:AWatering:ALine	9	58.9	0.5940	NA	NA	Nonsignificant
13	Rm_Temperature:ALine	9	58.9	0.5552	NA	NA	Nonsignificant
14	Rm_Temperature:AWatering	1	58.9	0.3359	NA	NA	Nonsignificant

Information criteria for spatial models

```
NULL

Chosen terms
[1] "Rm_Temperature:HLine"

LSDs
          c   minLSD  meanLSD   maxLSD assignedLSD accuracyLSD falsePos falseNeg
overall 780 4.526069 5.683925 6.616287    5.683925   0.2037072       12       22

Joining with `by = join_by(Rm_Temperature, Watering, Line.ID)`
Joining with `by = join_by(Watering, Line.ID, Rm_Temperature)`


#### Results for StomatalDensity


  Final model converged



#### Formulae from asreml object

fixed: StomatalDensity ~ ASide/ABlock + Rm_Temperature/(AWatering * ALine +
     HWatering * HLine)
random: NULL
residual: ~ (dsum(~ units | Rm_Temperature))




####  Summary of the fitted variance parameters for StomatalDensity

                           component std.error  z.ratio bound %ch
Rm_Temperature_7,Heat!R     39.48046  7.409081 5.328658     P   0
Rm_Temperature_8,Ambient!R  84.36620 15.552334 5.424665     P   0


####  Pseudo-anova table for fixed terms


Wald tests for fixed effects.
Response: StomatalDensity

                                Df denDF  F.inc  F.con Margin     Pr
(Intercept)                      1 101.6 7756.0 7756.0        0.0000
ASide                            1 101.6    2.2    2.0      B 0.1586
ASide:ABlock                     2 101.7    5.7    6.0      C 0.0035
Rm_Temperature                   1 104.0    0.1    0.1      A 0.7795
Rm_Temperature:AWatering         1  58.9    0.9    0.9      B 0.3359
Rm_Temperature:ALine             9  58.9    0.9    0.9      B 0.5552
Rm_Temperature:AWatering:ALine   9  58.9    0.8    0.8      C 0.5940
Rm_Temperature:HWatering         1  56.8    0.9    1.1      B 0.3067
Rm_Temperature:HLine             9  56.8    7.1    7.1      B 0.0000
Rm_Temperature:HWatering:HLine   9  56.8    1.7    1.7      C 0.1123
```

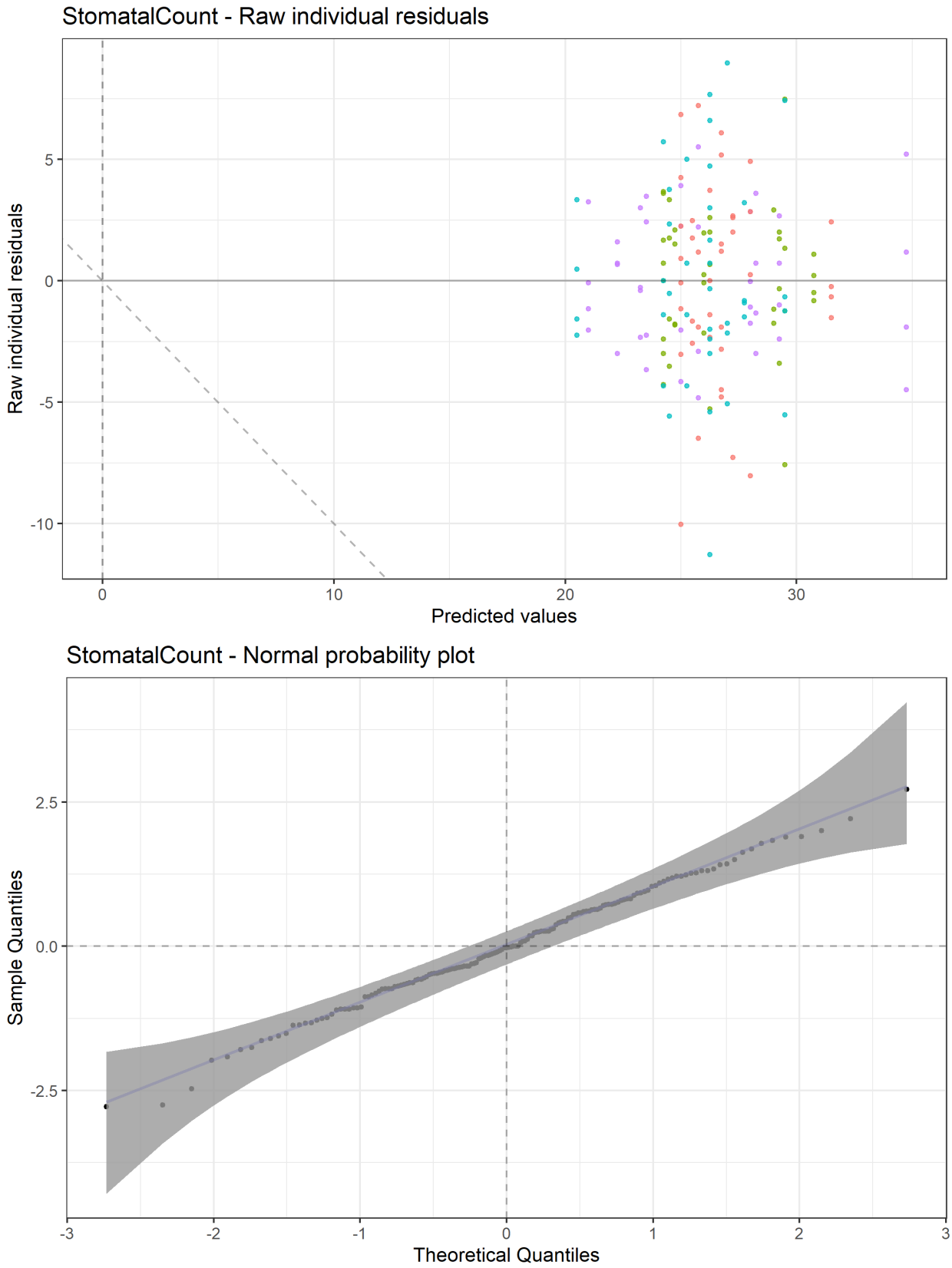


Figure 1: plot of chunk show-v2

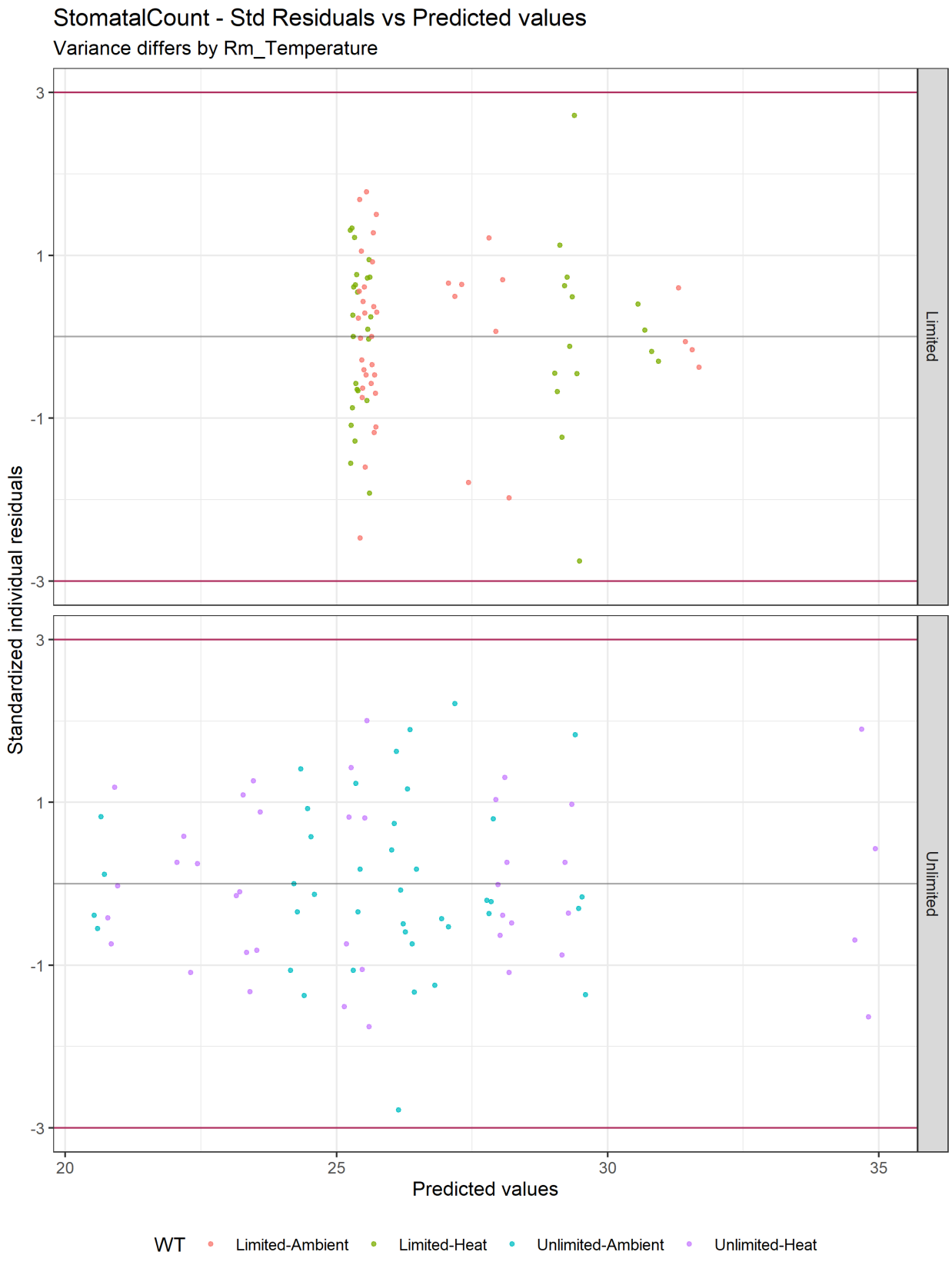


Figure 2: plot of chunk show-v2

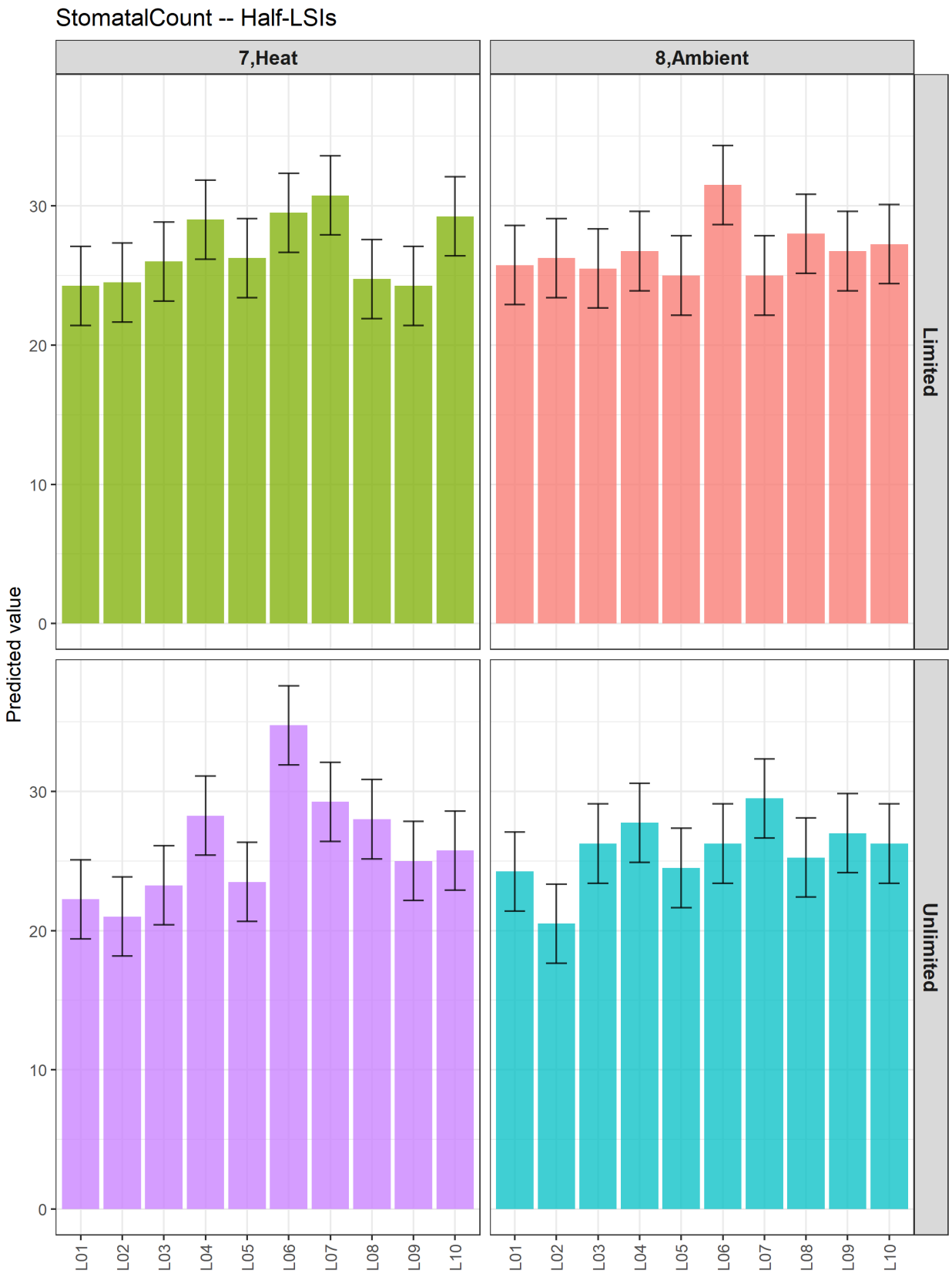


Figure 3: plot of chunk show-v2

```
####  Sequence of model investigations for StomatalDensity

(If a row has NA for p but not denDF, DF and denDF relate to fixed and variance parameter numbers)

                                                                                terms DF denDF      p
1                                                                       Initial model 43   1.0     NA
2  Start with residual heterogeneity in Watering & Rm_Temperature for Residual NA    NA     NA
3                                 Try homogeneity in Rm_Temperature for Residual  0  -2.0     NA
4                                       Try homogeneity in Watering for Residual  0  -2.0     NA
5                                                                    Room:Side  1    NA     NA
6                                                            Room:Side:Location  1    NA     NA
7                                                       Room:Side:Location:SubUnit  1    NA     NA
8                                                    Fix bound variance components  0   0.0     NA
9                                                   Rm_Temperature:HWatering:HLine  9  56.8 0.1123
10                                                            Rm_Temperature:HLine  9  56.8 0.0000
11                                                          Rm_Temperature:HWatering  1  56.8 0.3067
12                                                  Rm_Temperature:AWatering:ALine  9  58.9 0.5940
13                                                            Rm_Temperature:ALine  9  58.9 0.5552
14                                                          Rm_Temperature:AWatering  1  58.9 0.3359
          AIC        BIC           action
1  860.894764 995.926549   Starting model
2          NA         NA Changed residual
3    4.283150  -1.224031        Unswapped
4   -3.861354  -9.368534          Swapped
5          NA         NA         Boundary
6          NA         NA         Boundary
7          NA         NA         Boundary
8    0.000000   0.000000          Swapped
9          NA         NA   Nonsignificant
10         NA         NA      Significant
11         NA         NA   Nonsignificant
12         NA         NA   Nonsignificant
13         NA         NA   Nonsignificant
14         NA         NA   Nonsignificant

  Information criteria for spatial models

NULL

Chosen terms
[1] "Rm_Temperature:HLine"

LSDs
          c   minLSD  meanLSD   maxLSD assignedLSD accuracyLSD falsePos falseNeg
overall 780 8.799917 11.05111 12.86387    11.05111   0.2037072       12       22

Joining with `by = join_by(Rm_Temperature, Watering, Line.ID)`
Joining with `by = join_by(Watering, Line.ID, Rm_Temperature)`


#### Results for StomatalArea
```

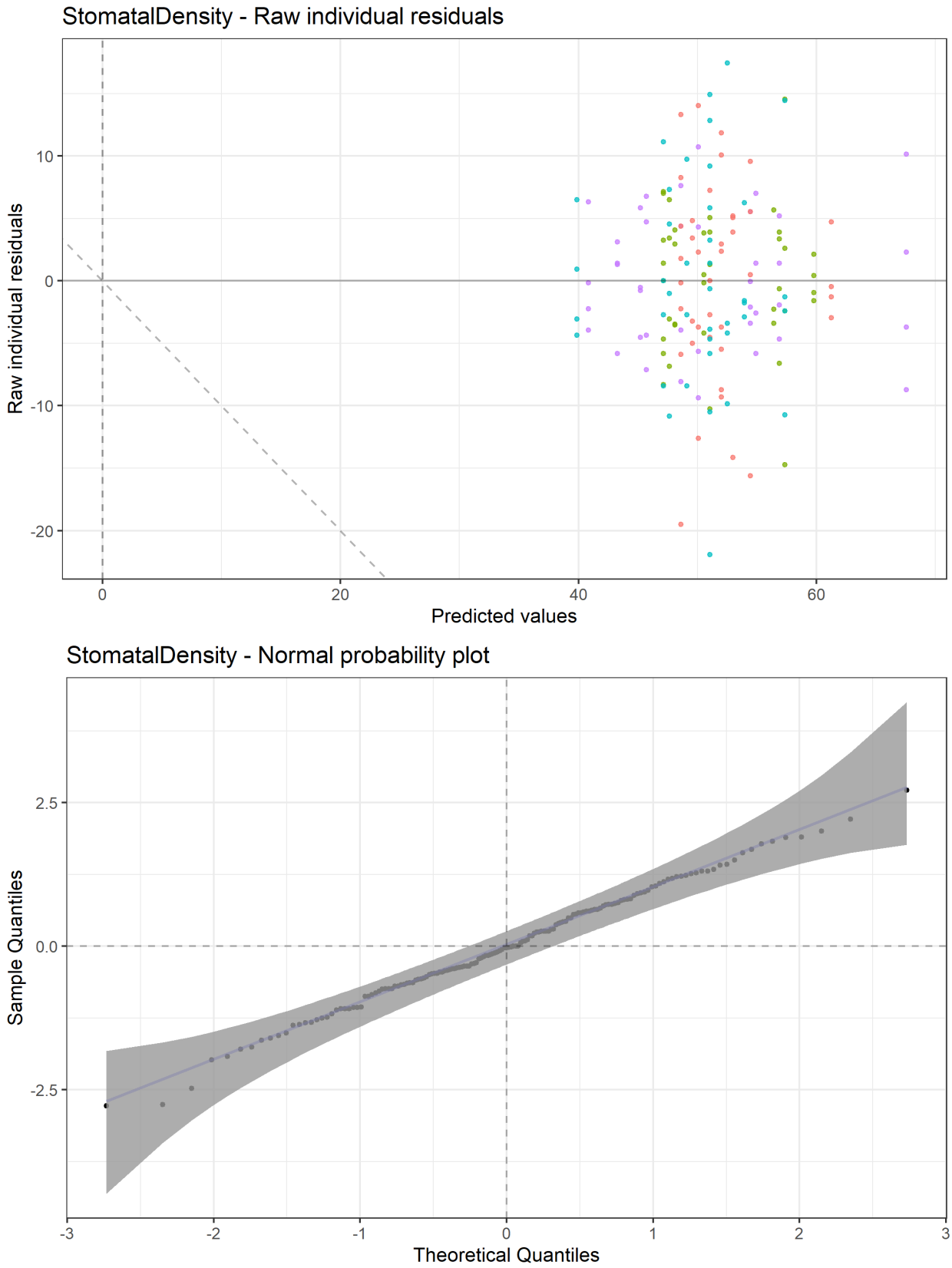


Figure 4: plot of chunk show-v2

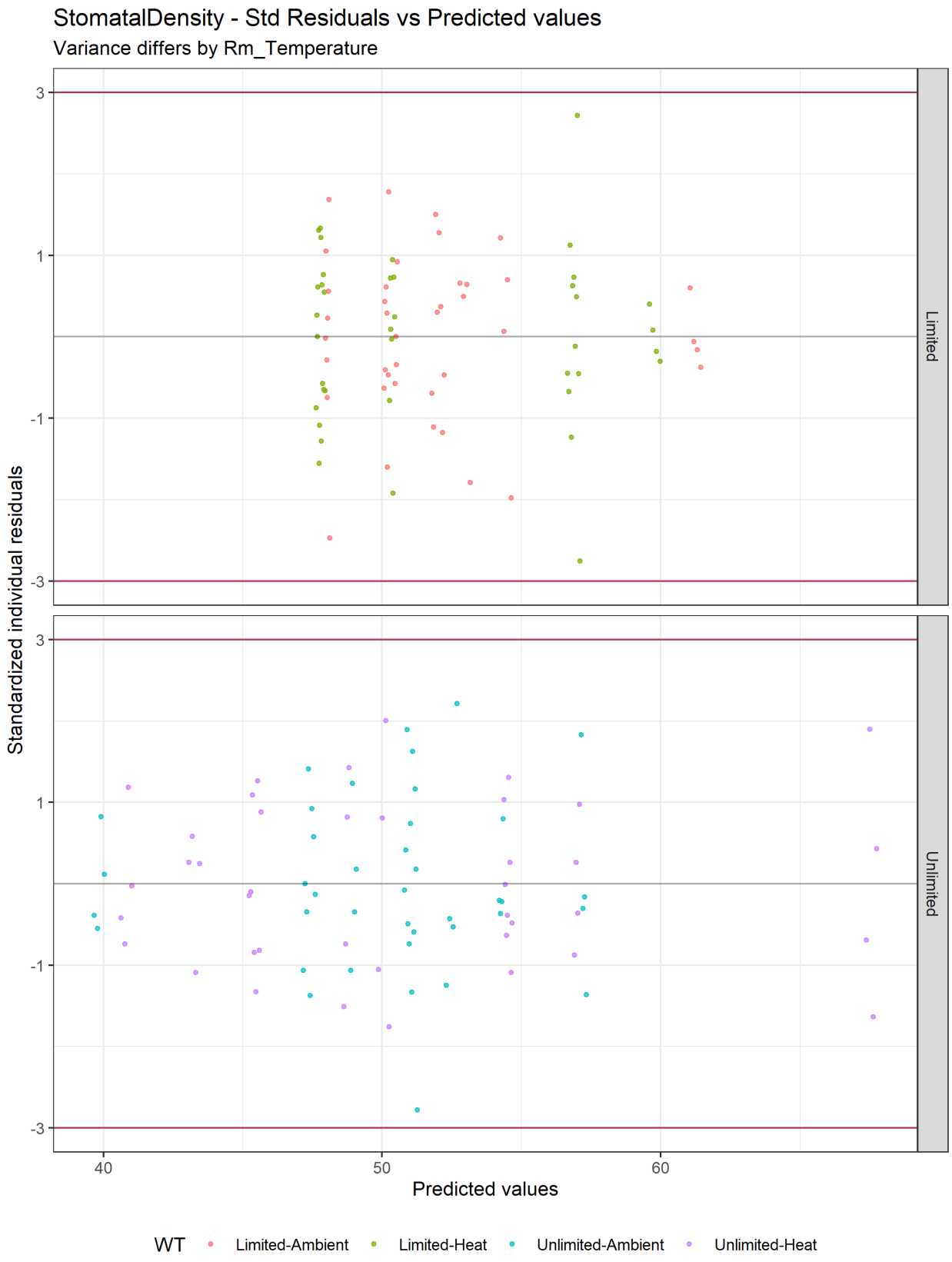


Figure 5: plot of chunk show-v2

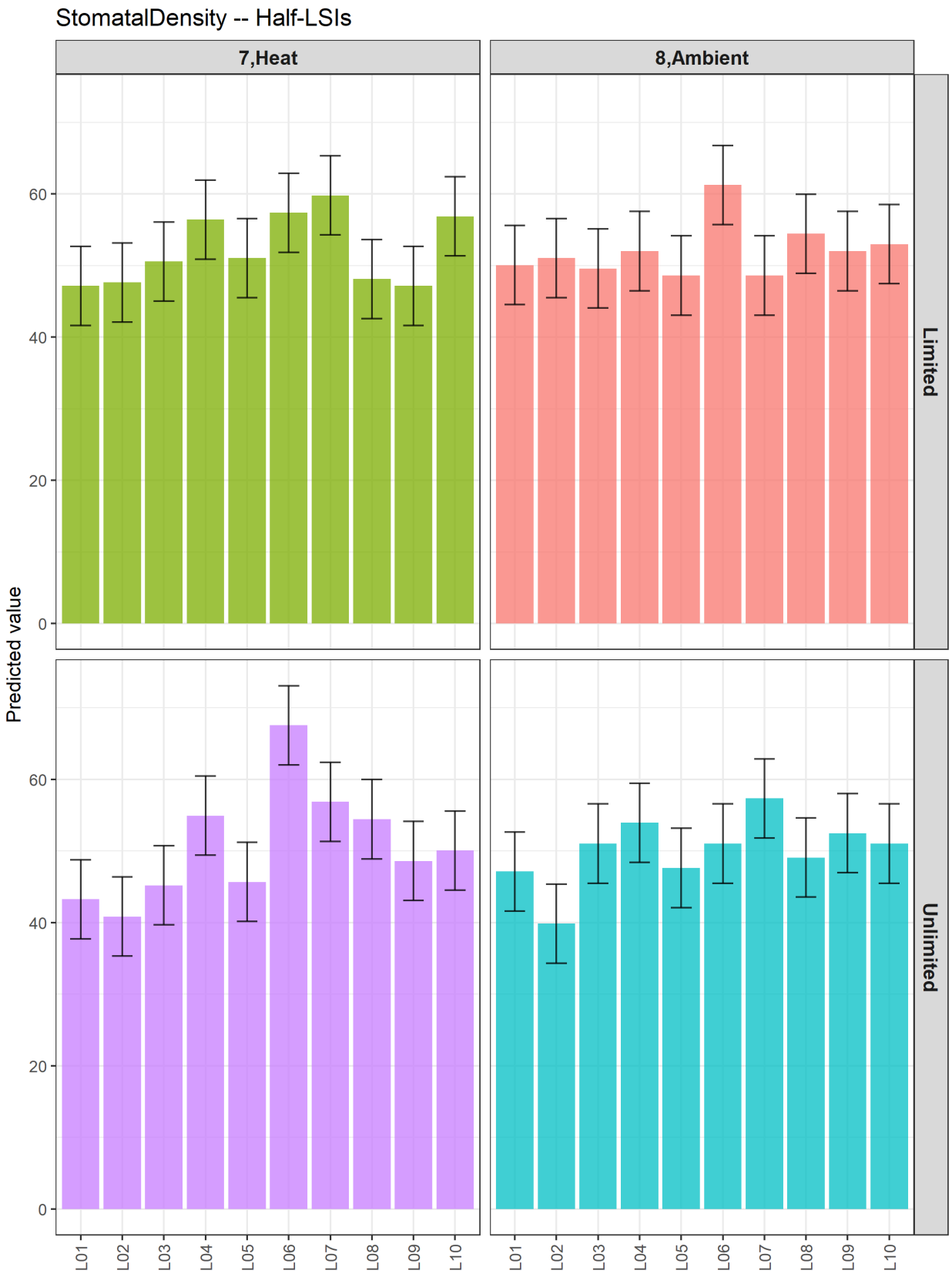


Figure 6: plot of chunk show-v2

```
  Final model converged


#### Formulae from asreml object

fixed: StomatalArea ~ ASide/ABlock + Rm_Temperature/(AWatering * ALine +
     HWatering * HLine)
random: ~ Room:Side:Location:SubUnit + Room:Side:Location:SubUnit:Cell
residual: ~ (dsum(~ units | Rm_Temperature))




####  Summary of the fitted variance parameters for StomatalArea

                                component std.error   z.ratio bound %ch
Room:Side:Location:SubUnit       4757.396  3744.469  1.270513     P   0
Room:Side:Location:SubUnit:Cell 28842.321  4263.616  6.764756     P   0
Rm_Temperature_7,Heat!R         62206.571  1965.546 31.648500     P   0
Rm_Temperature_8,Ambient!R      56309.328  1775.405 31.716335     P   0


####  Pseudo-anova table for fixed terms


Wald tests for fixed effects.
Response: StomatalArea

                               Df denDF  F.inc  F.con Margin     Pr
(Intercept)                     1   9.0 5406.0 5406.0        0.0000
ASide                           1   9.0    2.6    2.5      B 0.1488
ASide:ABlock                    2   9.0    0.0    0.0      C 0.9738
Rm_Temperature                  1   9.0    0.4    0.4      A 0.5564
Rm_Temperature:AWatering        1   8.9    2.6    2.6      B 0.1418
Rm_Temperature:ALine            9 106.1    3.4    3.4      B 0.0009
Rm_Temperature:AWatering:ALine  9 106.1    1.4    1.4      C 0.1784
Rm_Temperature:HWatering        1   9.1    1.4    1.3      B 0.2868
Rm_Temperature:HLine            9 107.6    3.4    3.4      B 0.0012
Rm_Temperature:HWatering:HLine  9 107.6    0.2    0.2      C 0.9863


####  Sequence of model investigations for StomatalArea

(If a row has NA for p but not denDF, DF and denDF relate to fixed and variance parameter numbers)

                                                                      terms DF denDF      p
1                                                                Initial model 43   3.0     NA
2  Start with residual heterogeneity in Watering & Rm_Temperature for Residual NA    NA     NA
3                              Try homogeneity in Rm_Temperature for Residual  0  -2.0     NA
4                                    Try homogeneity in Watering for Residual  0  -2.0     NA
5                                                                 Room:Side  1    NA     NA
6                                                          Room:Side:Location  1    NA     NA
```

```
7                                                  Fix bound variance components  0   0.0     NA
8                                                Rm_Temperature:HWatering:HLine  9 107.6 0.9863
9                                                          Rm_Temperature:HLine  9 107.6 0.0012
10                                                      Rm_Temperature:HWatering  1   9.1 0.2868
11                                               Rm_Temperature:AWatering:ALine  9 106.1 0.1784
12                                                         Rm_Temperature:ALine  9 106.1 0.0009
13                                                      Rm_Temperature:AWatering  1   8.9 0.1418
             AIC         BIC           action
1  50522.9683907 50814.45337   Starting model
2             NA          NA Changed residual
3      0.9281318   -11.72442        Unswapped
4     -1.5501424   -14.20269          Swapped
5             NA          NA         Boundary
6             NA          NA         Boundary
7      0.0000000     0.00000          Swapped
8             NA          NA   Nonsignificant
9             NA          NA      Significant
10            NA          NA   Nonsignificant
11            NA          NA   Nonsignificant
12            NA          NA      Significant
13            NA          NA   Nonsignificant

  Information criteria for spatial models

NULL

Chosen terms
[1] "Rm_Temperature:HLine" "Rm_Temperature:ALine"

LSDs
          c   minLSD  meanLSD   maxLSD assignedLSD accuracyLSD falsePos falseNeg
overall 780 243.1809 259.7307 283.2847    259.7307  0.09068625        5        4

Joining with `by = join_by(Rm_Temperature, Watering, Line.ID)`
Joining with `by = join_by(Watering, Line.ID, Rm_Temperature)`



#### Results for StomatalWidth


  Final model converged



#### Formulae from asreml object

fixed: StomatalWidth ~ ASide/ABlock + Rm_Temperature/(AWatering * ALine +
     HWatering * HLine)
random: ~ Room:Side + Room:Side:Location:SubUnit +
     Room:Side:Location:SubUnit:Cell
residual: ~ (dsum(~ units | Watering))
```

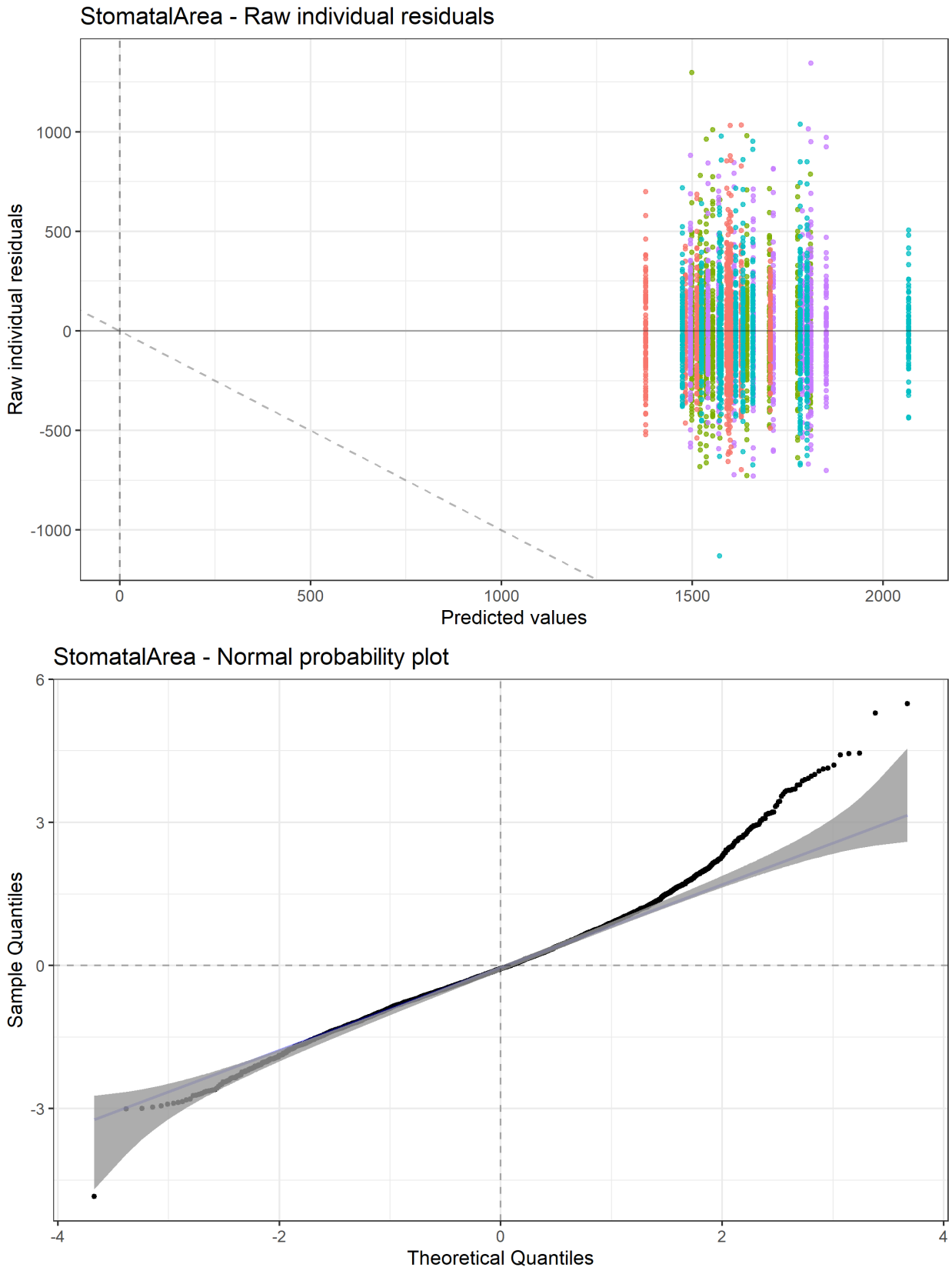


Figure 7: plot of chunk show-v2

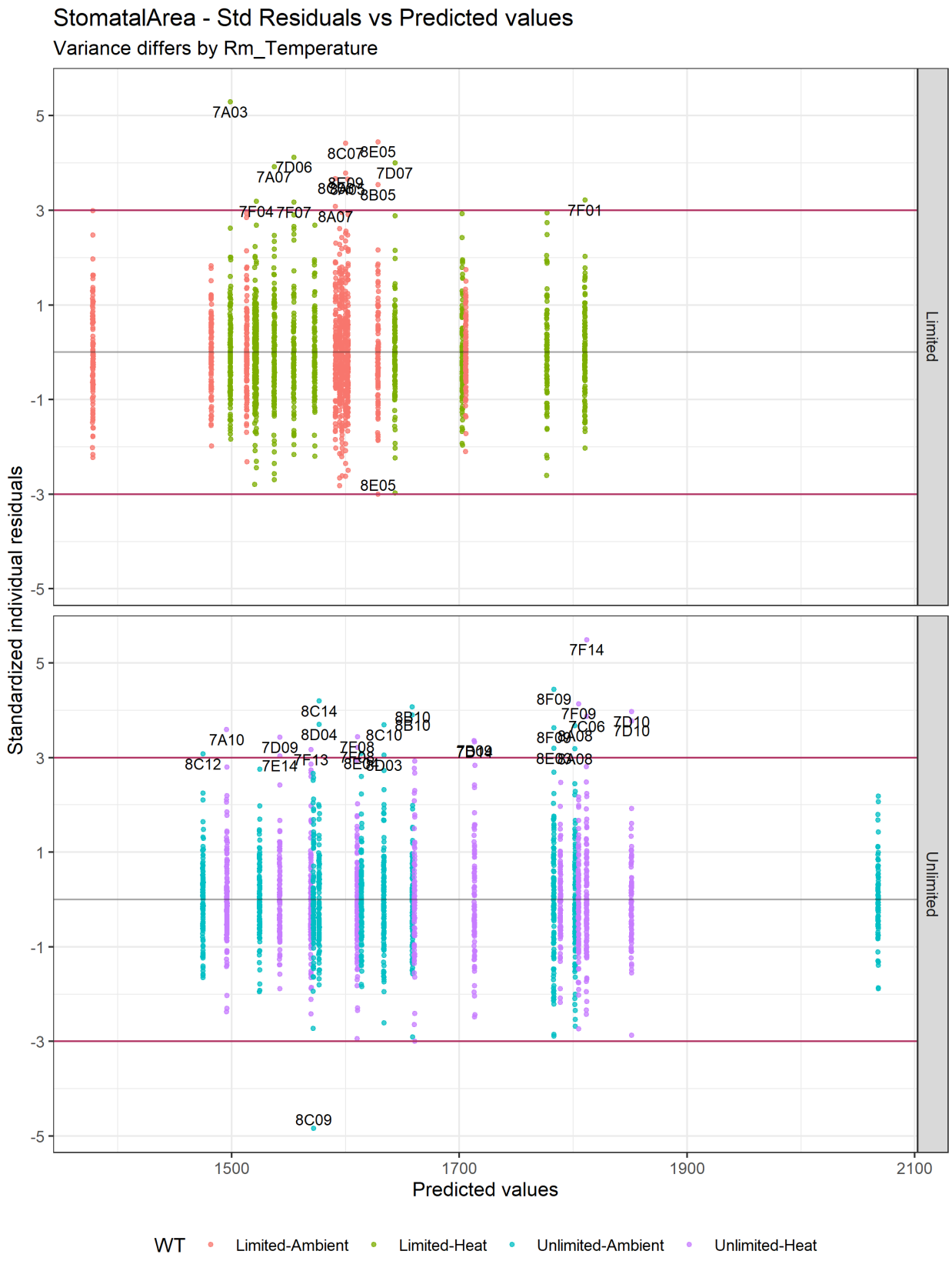


Figure 8: plot of chunk show-v2

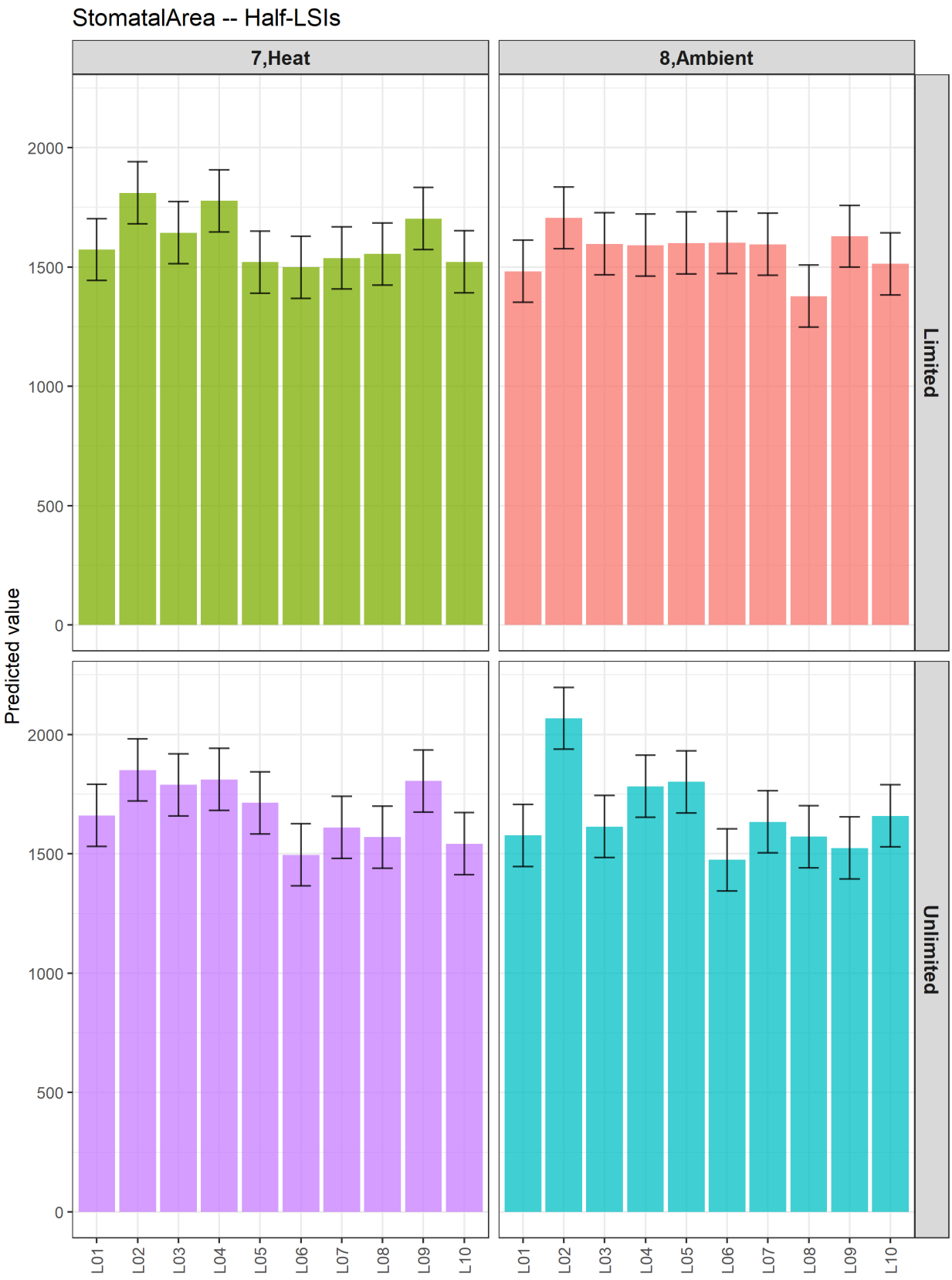


Figure 9: plot of chunk show-v2

```
####  Summary of the fitted variance parameters for StomatalWidth

                                 component std.error    z.ratio bound %ch
Room:Side                        0.1308372 0.6039463  0.2166371     P   0
Room:Side:Location:SubUnit       0.2068712 0.5808275  0.3561664     P   0
Room:Side:Location:SubUnit:Cell  8.5678985 1.2597242  6.8014080     P   0
Watering_Limited!R              15.2113739 0.4768344 31.9007489     P   0
Watering_Unlimited!R            17.1399116 0.5434500 31.5390799     P   0


####  Pseudo-anova table for fixed terms


Wald tests for fixed effects.
Response: StomatalWidth

                               Df denDF   F.inc   F.con Margin     Pr
(Intercept)                     1   1.0 11260.0 11260.0        0.0060
ASide                           1   1.0     2.4     2.2      B 0.3759
ASide:ABlock                    2   8.0     0.1     0.1      C 0.9008
Rm_Temperature                  1   1.0     0.3     0.3      A 0.6993
Rm_Temperature:AWatering        1   7.9     7.6     7.6      B 0.0248
Rm_Temperature:ALine            9 106.7     2.4     2.4      B 0.0153
Rm_Temperature:AWatering:ALine  9 106.8     2.3     2.3      C 0.0214
Rm_Temperature:HWatering        1   8.1     6.7     6.0      B 0.0397
Rm_Temperature:HLine            9 106.8     3.1     3.1      B 0.0025
Rm_Temperature:HWatering:HLine  9 106.8     0.2     0.2      C 0.9888


####  Sequence of model investigations for StomatalWidth

(If a row has NA for p but not denDF, DF and denDF relate to fixed and variance parameter numbers)

                                                                                 terms DF denDF      p
1                                                                      Initial model 43   4.0     NA
2  Start with residual heterogeneity in Watering & Rm_Temperature for Residual NA    NA     NA
3                                Try homogeneity in Rm_Temperature for Residual  0  -2.0     NA
4                     Try homogeneity in Watering & Rm_Temperature for Residual  0  -1.0     NA
5                                                             Room:Side:Location  1    NA     NA
6                                              Fix bound variance components  0   0.0     NA
7                                             Rm_Temperature:HWatering:HLine  9 106.8 0.9888
8                                                      Rm_Temperature:HLine  9 106.8 0.0025
9                                                 Rm_Temperature:HWatering  1   8.1 0.0397
10                                            Rm_Temperature:AWatering:ALine  9 106.8 0.0214
            AIC         BIC           action
1  16311.353418 16609.28750   Starting model
2            NA          NA Changed residual
3     -2.227427   -14.88481          Swapped
4      5.165772    -1.16292        Unswapped
5            NA          NA         Boundary
6      0.000000     0.00000          Swapped
7            NA          NA   Nonsignificant
8            NA          NA      Significant
```

```
9            NA          NA      Significant
10           NA          NA      Significant

  Information criteria for spatial models

NULL

Chosen terms
[1] "Rm_Temperature:HLine"           "Rm_Temperature:HWatering"
[3] "Rm_Temperature:AWatering:ALine"

LSDs
          c   minLSD  meanLSD   maxLSD assignedLSD accuracyLSD falsePos falseNeg
overall 780 4.177098 4.289428 4.661469    4.289428  0.08673447        5        2

Joining with `by = join_by(Rm_Temperature, Watering, Line.ID)`
Joining with `by = join_by(Watering, Line.ID, Rm_Temperature)`


#### Results for StomatalLength


  Final model converged



#### Formulae from asreml object

fixed: StomatalLength ~ ASide/ABlock + Rm_Temperature/(AWatering * ALine +
     HWatering * HLine)
random: ~ Room:Side:Location:SubUnit + Room:Side:Location:SubUnit:Cell
residual: ~ (dsum(~ units | Watering))




####  Summary of the fitted variance parameters for StomatalLength

                                component std.error   z.ratio bound %ch
Room:Side:Location:SubUnit       1.327791 1.5819205  0.839354     P   0
Room:Side:Location:SubUnit:Cell 18.421640 2.6845906  6.861992     P   0
Watering_Limited!R              31.823611 0.9982471 31.879492     P   0
Watering_Unlimited!R            29.839456 0.9463076 31.532511     P   0


####  Pseudo-anova table for fixed terms


Wald tests for fixed effects.
Response: StomatalLength

                             Df denDF   F.inc   F.con Margin     Pr
(Intercept)                   1   9.0 18760.0 18760.0        0.0000
ASide                         1   9.0     0.3     0.4      B 0.5666
```

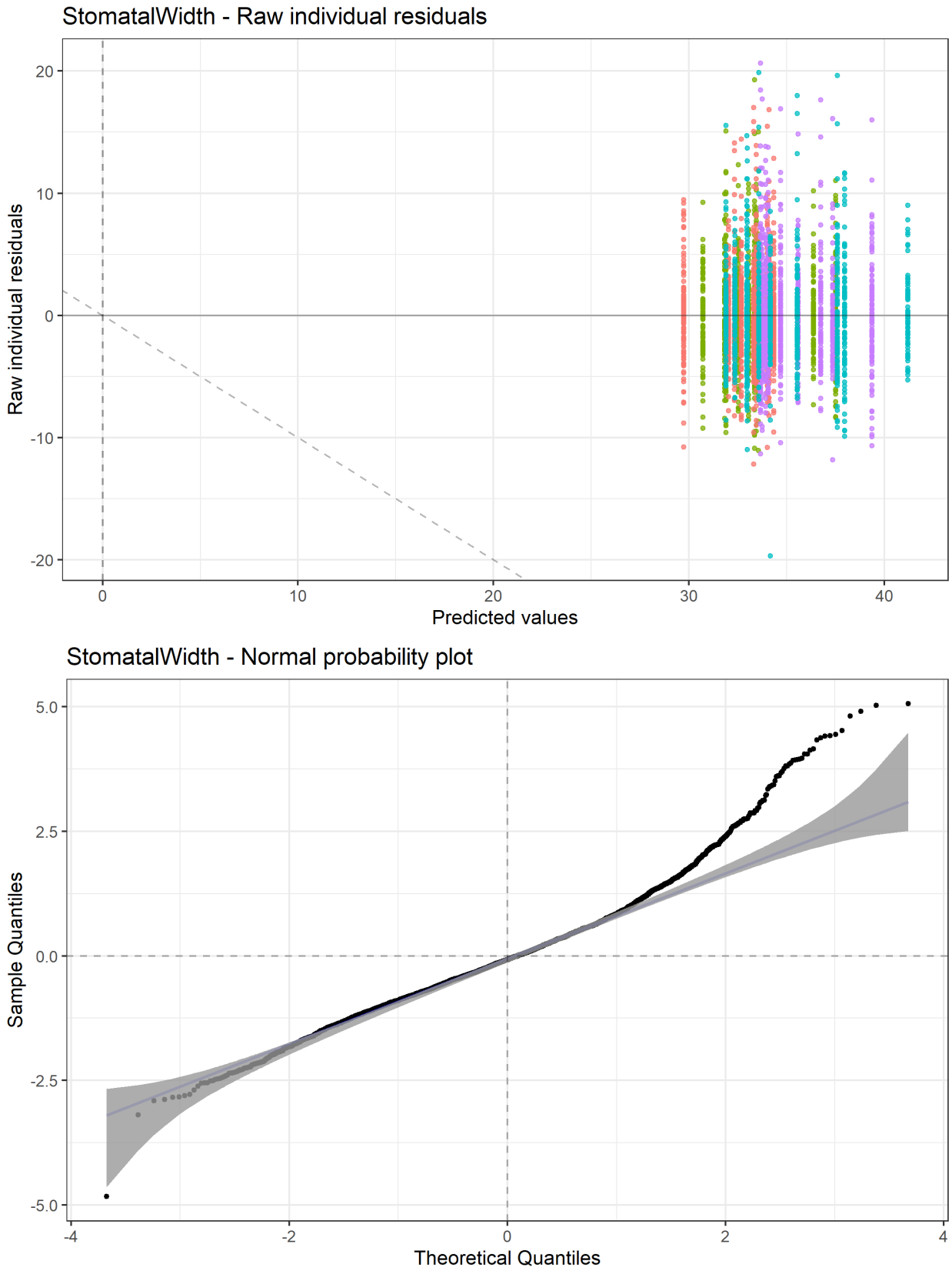


Figure 10: plot of chunk show-v2

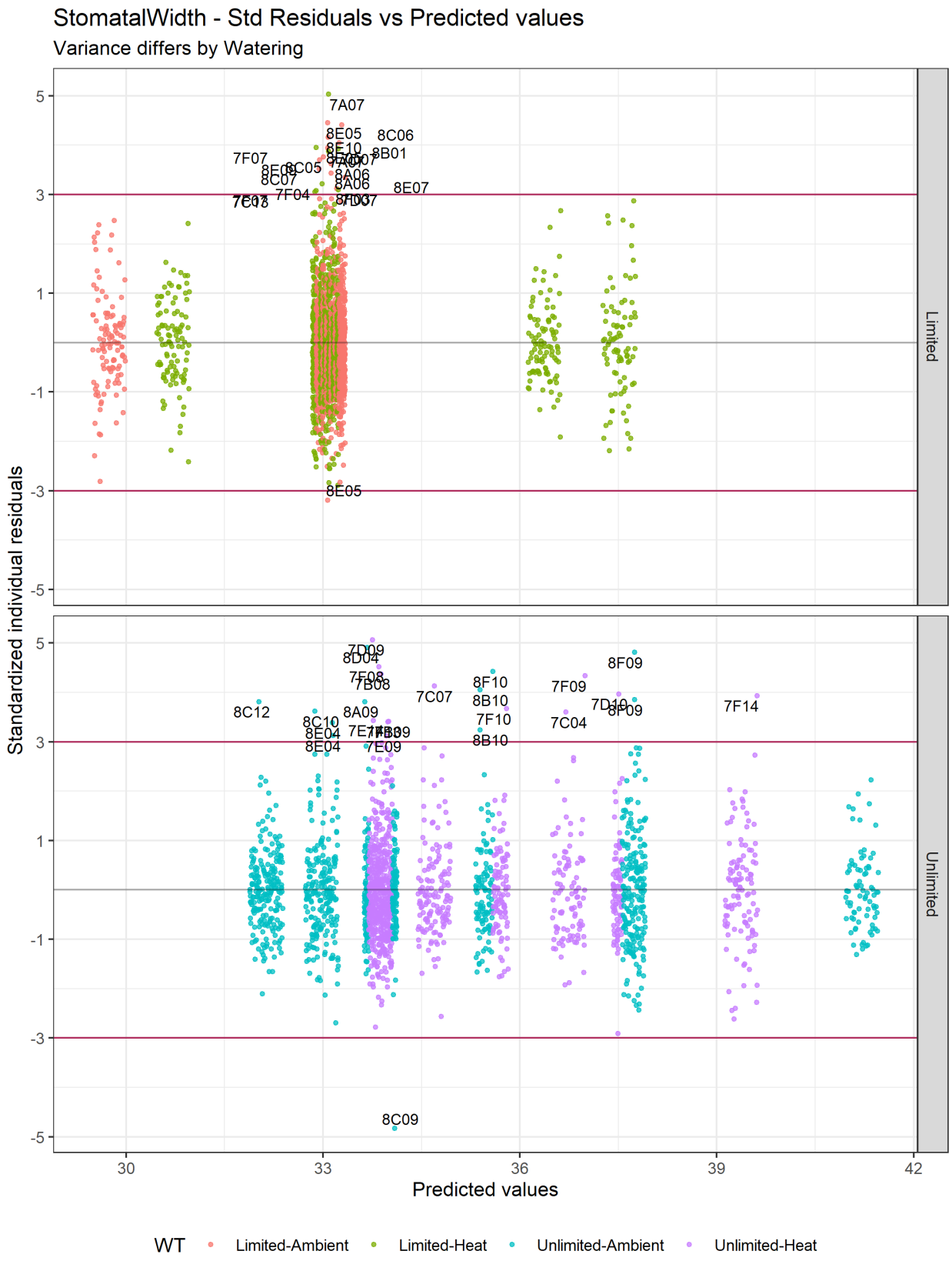


Figure 11: plot of chunk show-v2

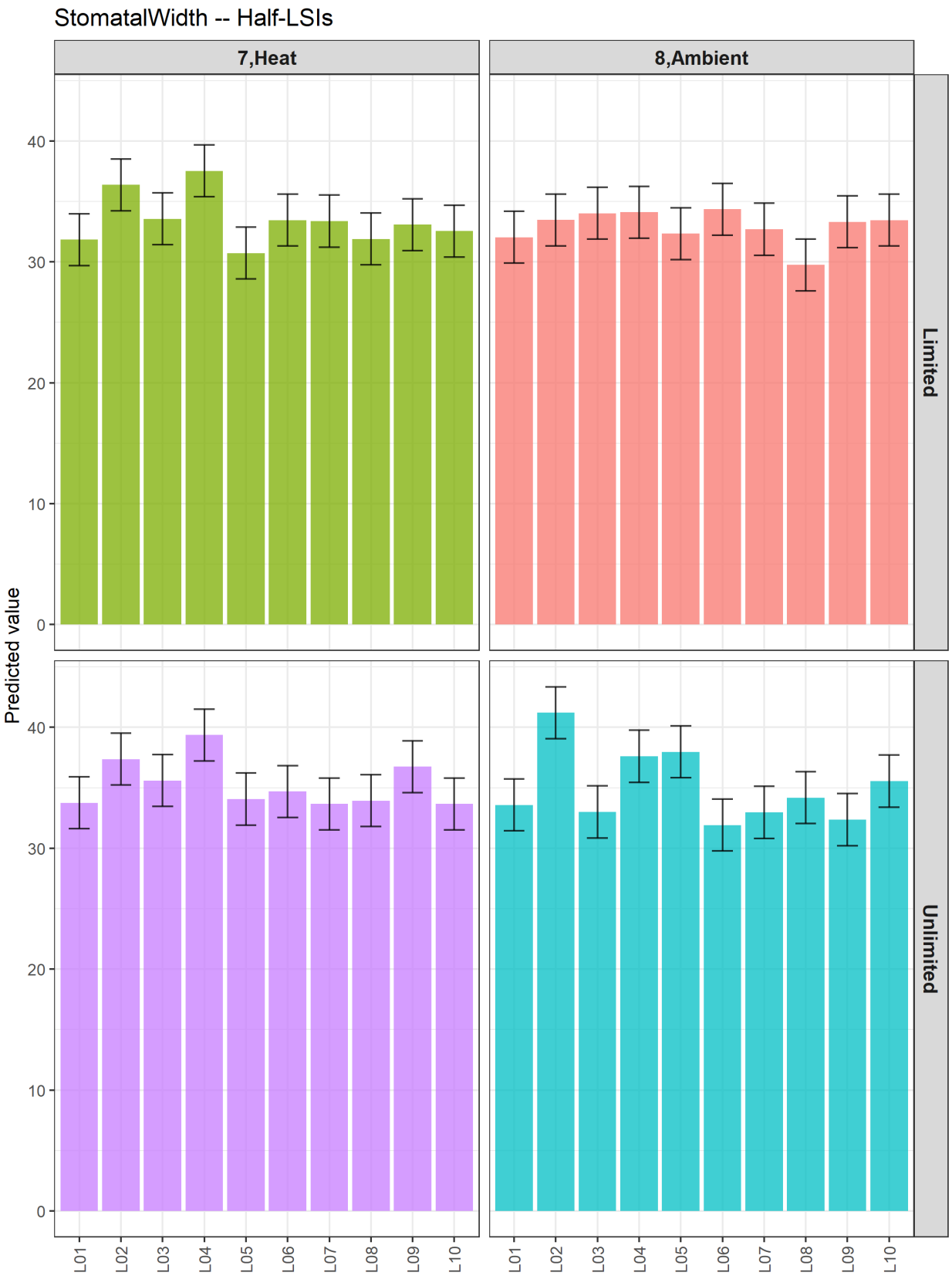


Figure 12: plot of chunk show-v2

```
ASide:ABlock                    2   9.0     0.6     0.6      C 0.5814
Rm_Temperature                  1   9.0     0.2     0.2      A 0.6586
Rm_Temperature:AWatering        1   8.9     0.0     0.0      B 0.8342
Rm_Temperature:ALine            9 107.0     1.5     1.5      B 0.1642
Rm_Temperature:AWatering:ALine  9 107.0     0.5     0.5      C 0.8761
Rm_Temperature:HWatering        1   9.1     0.1     0.1      B 0.7569
Rm_Temperature:HLine            9 106.9     3.6     3.6      B 0.0007
Rm_Temperature:HWatering:HLine  9 107.0     0.4     0.4      C 0.9060


####  Sequence of model investigations for StomatalLength

(If a row has NA for p but not denDF, DF and denDF relate to fixed and variance parameter numbers)

                                                                                terms DF denDF      p
1                                                                     Initial model 43   3.0     NA
2  Start with residual heterogeneity in Watering & Rm_Temperature for Residual NA    NA     NA
3                               Try homogeneity in Rm_Temperature for Residual  0  -2.0     NA
4                    Try homogeneity in Watering & Rm_Temperature for Residual  0  -1.0     NA
5                                                                   Room:Side  1    NA     NA
6                                                            Room:Side:Location  1    NA     NA
7                                                     Fix bound variance components  0   0.0     NA
8                                                   Rm_Temperature:HWatering:HLine  9 107.0 0.9060
9                                                             Rm_Temperature:HLine  9 106.9 0.0007
10                                                         Rm_Temperature:HWatering  1   9.1 0.7569
11                                                  Rm_Temperature:AWatering:ALine  9 107.0 0.8761
12                                                            Rm_Temperature:ALine  9 107.0 0.1642
13                                                         Rm_Temperature:AWatering  1   8.9 0.8342
             AIC          BIC           action
1   1.901561e+04 19307.161368   Starting model
2             NA           NA Changed residual
3  -1.454771e-01   -12.800929          Swapped
4   8.225861e-02    -6.245468        Unswapped
5             NA           NA         Boundary
6             NA           NA         Boundary
7   0.000000e+00     0.000000          Swapped
8             NA           NA   Nonsignificant
9             NA           NA      Significant
10            NA           NA   Nonsignificant
11            NA           NA   Nonsignificant
12            NA           NA   Nonsignificant
13            NA           NA   Nonsignificant

  Information criteria for spatial models

NULL

Chosen terms
[1] "Rm_Temperature:HLine"

LSDs
          c   minLSD  meanLSD   maxLSD assignedLSD accuracyLSD falsePos falseNeg
overall 780 6.101866 6.325752 6.860249    6.325752   0.0844955        0        2
```

```
Joining with `by = join_by(Rm_Temperature, Watering, Line.ID)`
Joining with `by = join_by(Watering, Line.ID, Rm_Temperature)`


#### Results for PoreArea


  Final model converged



#### Formulae from asreml object

fixed: PoreArea ~ ASide/ABlock + Rm_Temperature/(AWatering * ALine + HWatering *
     HLine)
random: ~ Room:Side:Location + Room:Side:Location:SubUnit +
     Room:Side:Location:SubUnit:Cell
residual: ~ (dsum(~ units | Watering))




####  Summary of the fitted variance parameters for PoreArea

                                 component std.error      z.ratio bound %ch
Room:Side:Location                11.36383 1283.0050  0.008857201     P   0
Room:Side:Location:SubUnit       982.86895 1489.8976  0.659688943     P   0
Room:Side:Location:SubUnit:Cell 14587.83278 2134.1613  6.835393833     P   0
Watering_Limited!R              26805.54901  840.8658 31.878509641     P   0
Watering_Unlimited!R            24833.75214  787.5466 31.533057599     P   0


####  Pseudo-anova table for fixed terms


Wald tests for fixed effects.
Response: PoreArea

                              Df denDF  F.inc  F.con Margin     Pr
(Intercept)                    1   3.0 5132.0 5132.0        0.0000
ASide                          1   3.0    0.4    0.4      B 0.5720
ASide:ABlock                   2   3.0    0.5    0.5      C 0.6405
Rm_Temperature                 1   3.0    0.2    0.2      A 0.6993
Rm_Temperature:AWatering       1   6.0    0.0    0.0      B 0.8637
Rm_Temperature:ALine           9 106.9    1.4    1.4      B 0.1946
Rm_Temperature:AWatering:ALine 9 106.9    0.6    0.6      C 0.8265
Rm_Temperature:HWatering       1   6.1    0.2    0.1      B 0.7376
Rm_Temperature:HLine           9 106.9    3.5    3.5      B 0.0007
Rm_Temperature:HWatering:HLine 9 106.9    0.5    0.5      C 0.8652


####  Sequence of model investigations for PoreArea
```

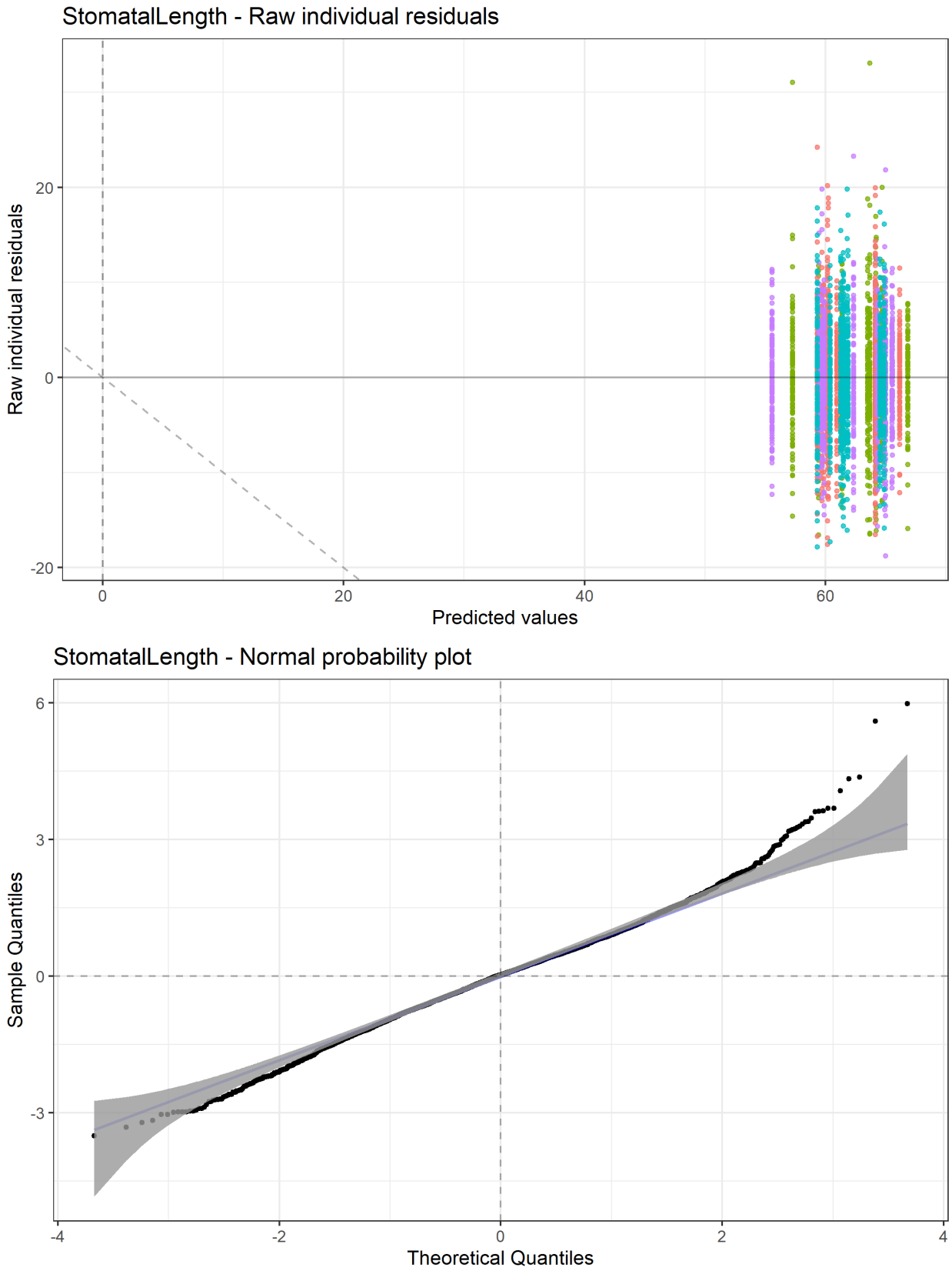


Figure 13: plot of chunk show-v2

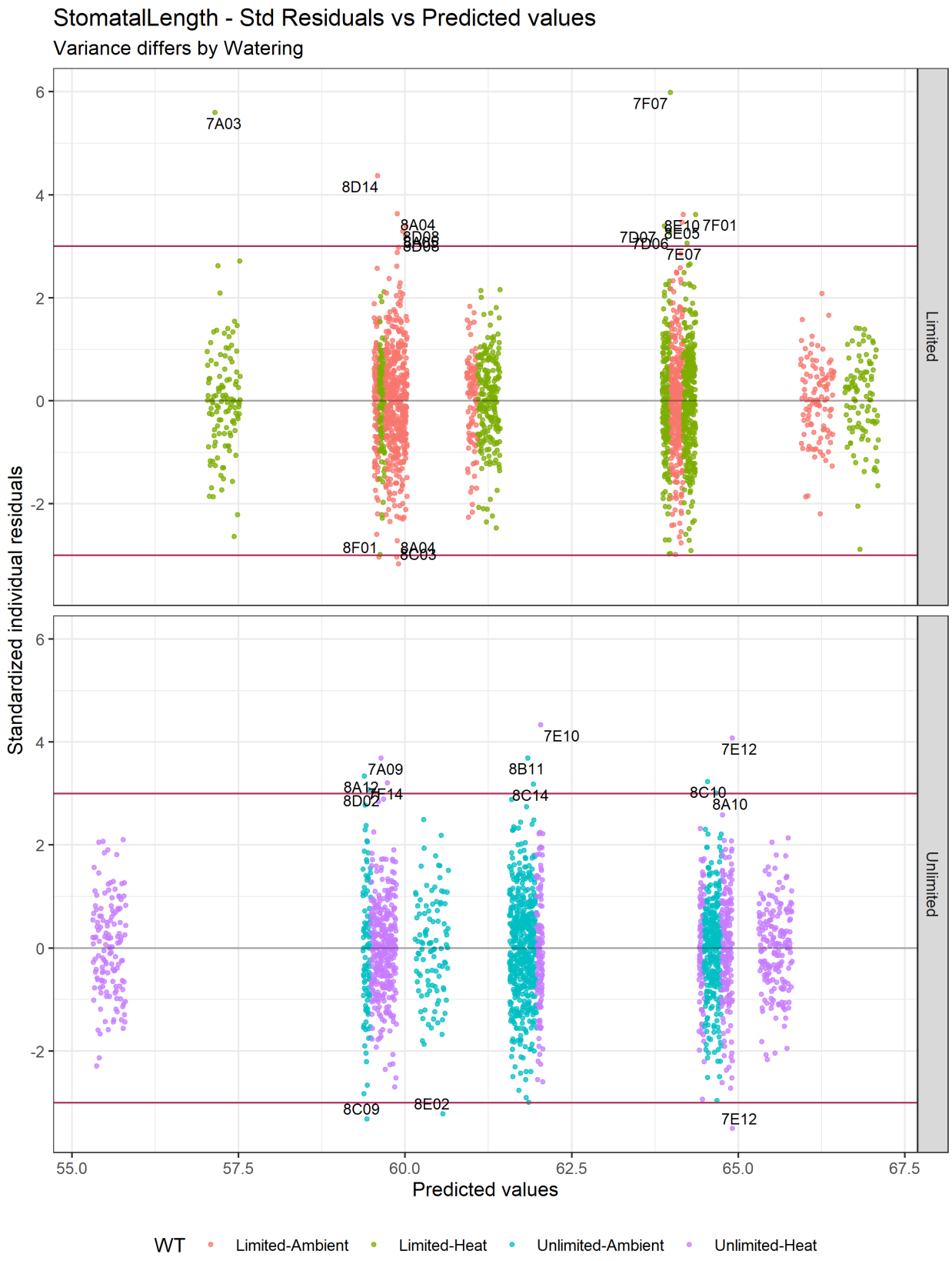


Figure 14: plot of chunk show-v2

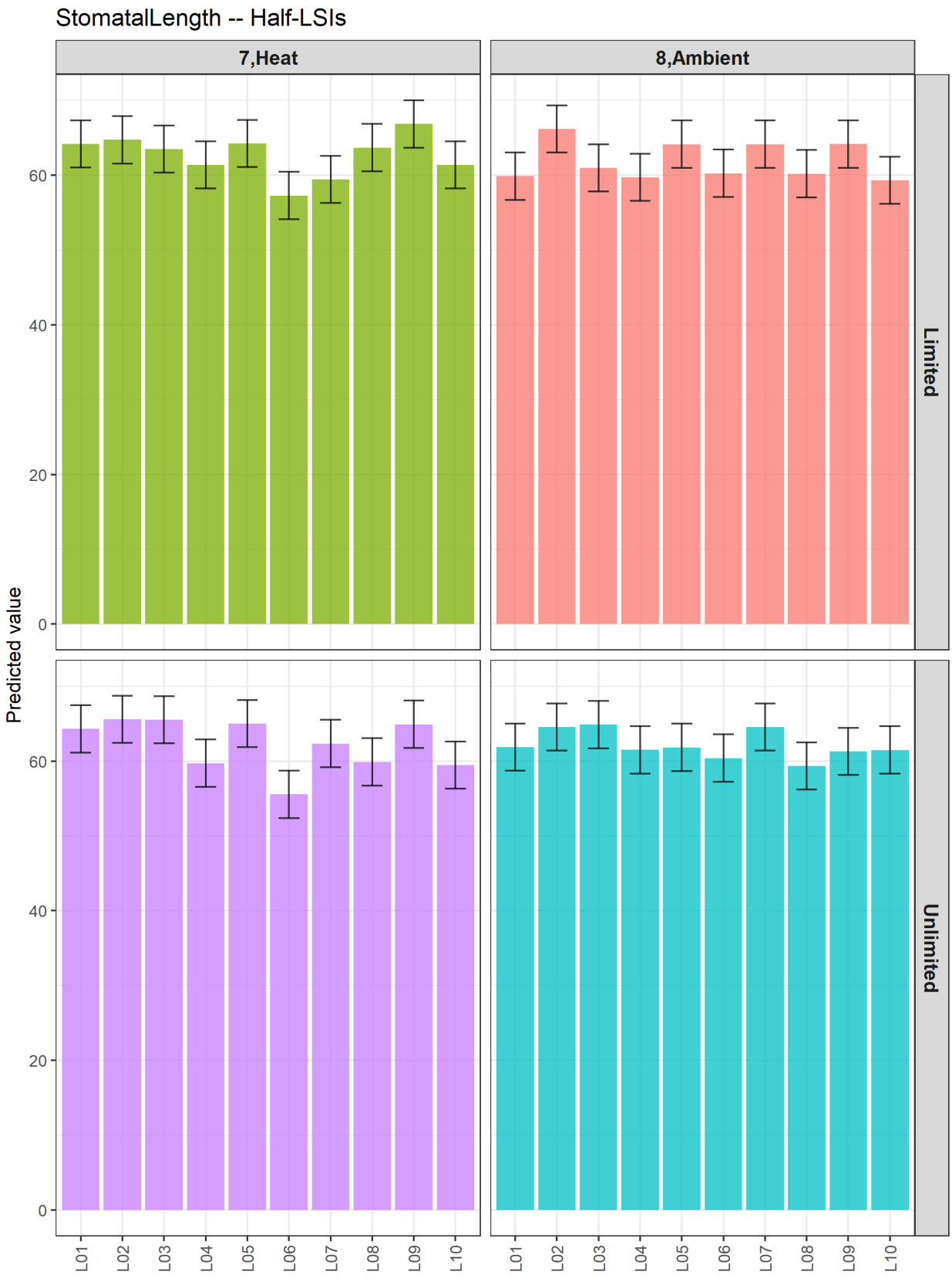


Figure 15: plot of chunk show-v2

```
(If a row has NA for p but not denDF, DF and denDF relate to fixed and variance parameter numbers)

                                                                                terms DF denDF      p
1                                                                      Initial model 43   4.0     NA
2  Start with residual heterogeneity in Watering & Rm_Temperature for Residual NA    NA     NA
3                                Try homogeneity in Rm_Temperature for Residual  0  -2.0     NA
4                     Try homogeneity in Watering & Rm_Temperature for Residual  0  -1.0     NA
5                                                                    Room:Side  1    NA     NA
6                                             Fix bound variance components  0   0.0     NA
7                                           Rm_Temperature:HWatering:HLine  9 106.9 0.8652
8                                                    Rm_Temperature:HLine  9 106.9 0.0007
9                                                Rm_Temperature:HWatering  1   6.1 0.7376
10                                          Rm_Temperature:AWatering:ALine  9 106.9 0.8265
11                                                   Rm_Temperature:ALine  9 106.9 0.1946
12                                               Rm_Temperature:AWatering  1   6.0 0.8637
             AIC          BIC           action
1  47142.0576204 47439.946747   Starting model
2             NA           NA Changed residual
3     -1.8820831   -14.537535          Swapped
4      0.9685117    -5.359214        Unswapped
5             NA           NA         Boundary
6      0.0000000     0.000000          Swapped
7             NA           NA   Nonsignificant
8             NA           NA      Significant
9             NA           NA   Nonsignificant
10            NA           NA   Nonsignificant
11            NA           NA   Nonsignificant
12            NA           NA   Nonsignificant

  Information criteria for spatial models

NULL

Chosen terms
[1] "Rm_Temperature:HLine"

LSDs
          c  minLSD  meanLSD   maxLSD assignedLSD accuracyLSD falsePos falseNeg
overall 780 171.924 178.0534 193.0861    178.0534   0.0844282        0        0

Joining with `by = join_by(Rm_Temperature, Watering, Line.ID)`
Joining with `by = join_by(Watering, Line.ID, Rm_Temperature)`


#### Results for gsmax


  Final model converged



#### Formulae from asreml object

fixed: gsmax ~ ASide/ABlock + Rm_Temperature/(AWatering * ALine + HWatering *
```

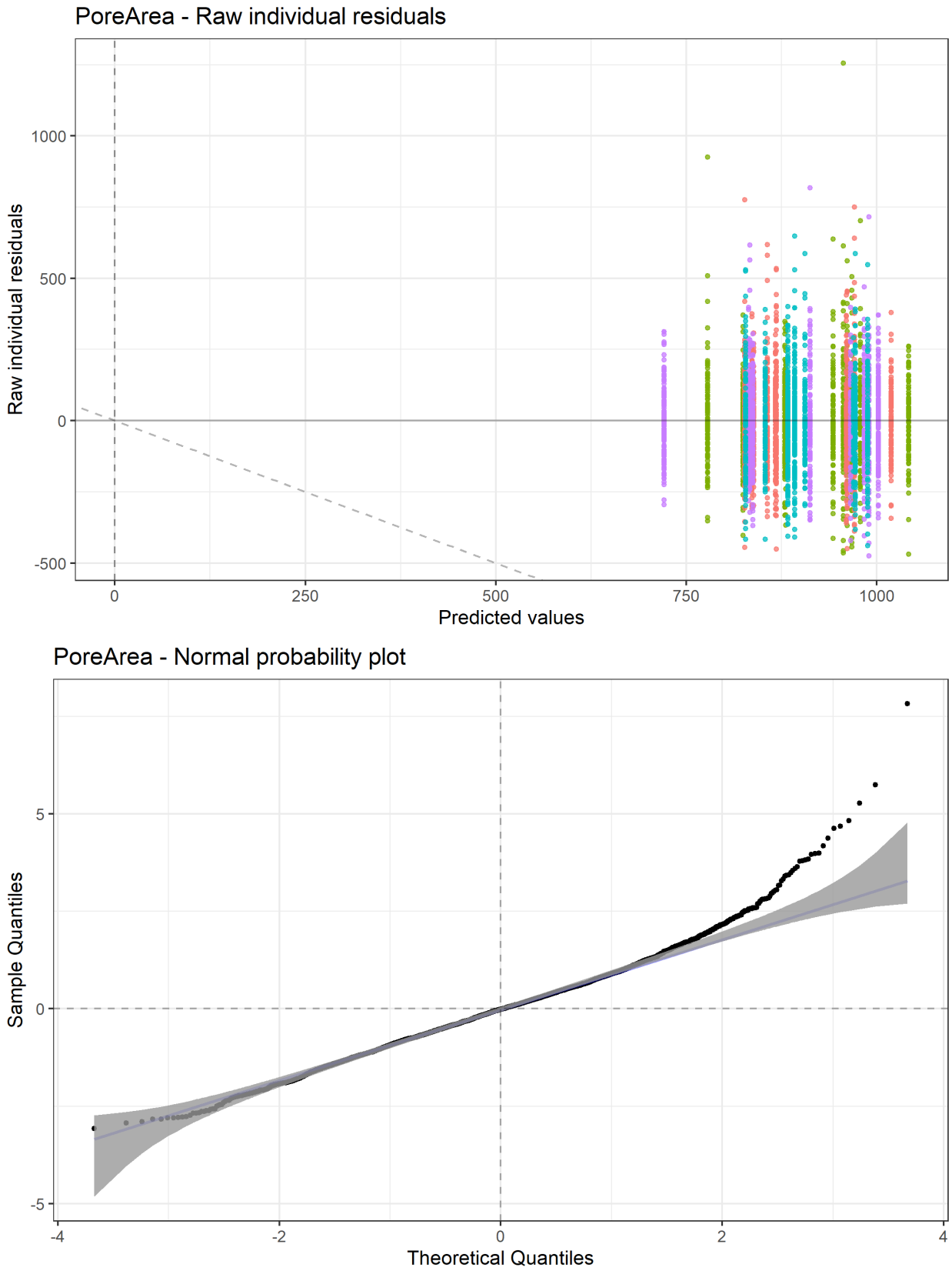


Figure 16: plot of chunk show-v2

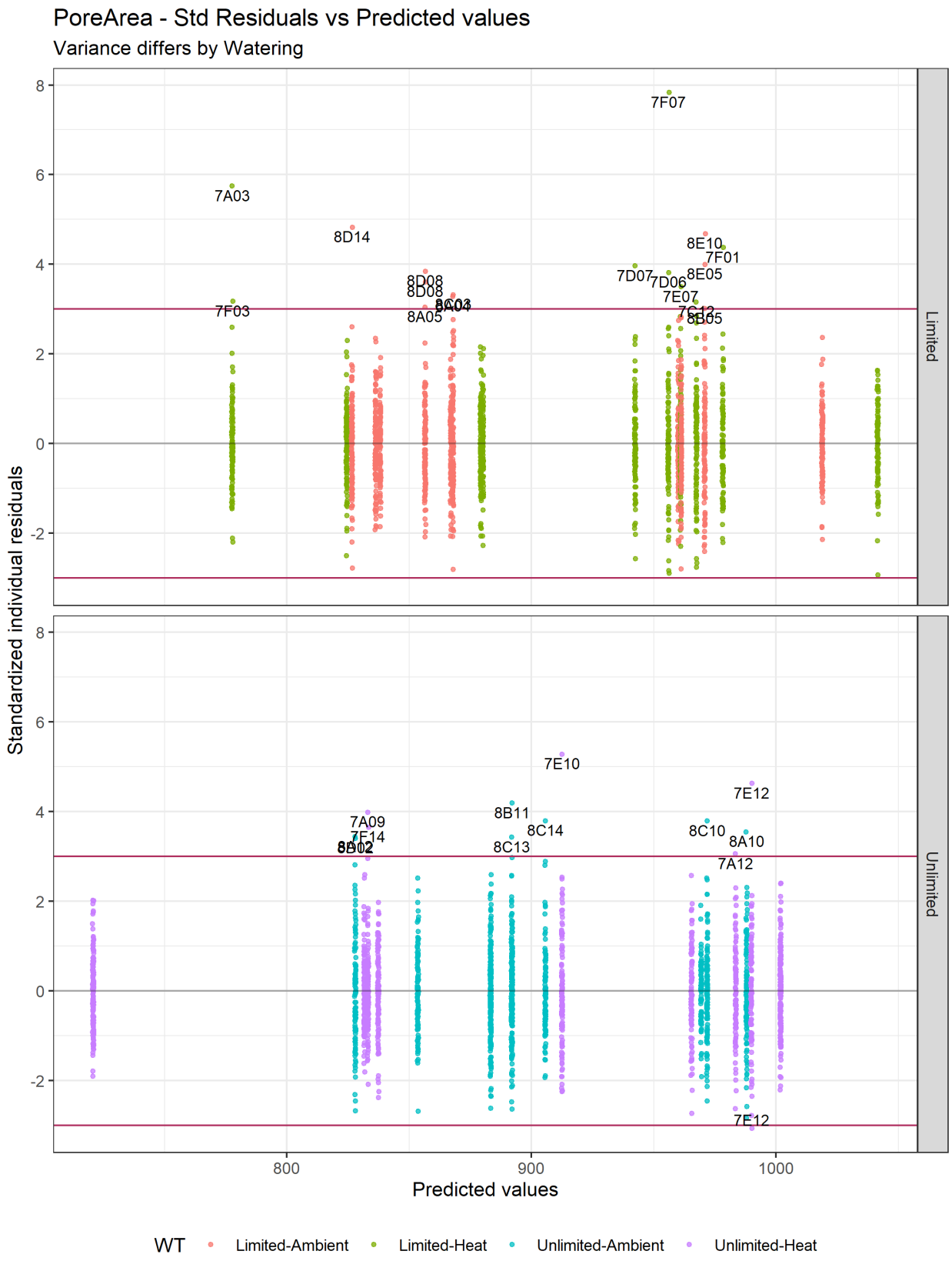


Figure 17: plot of chunk show-v2

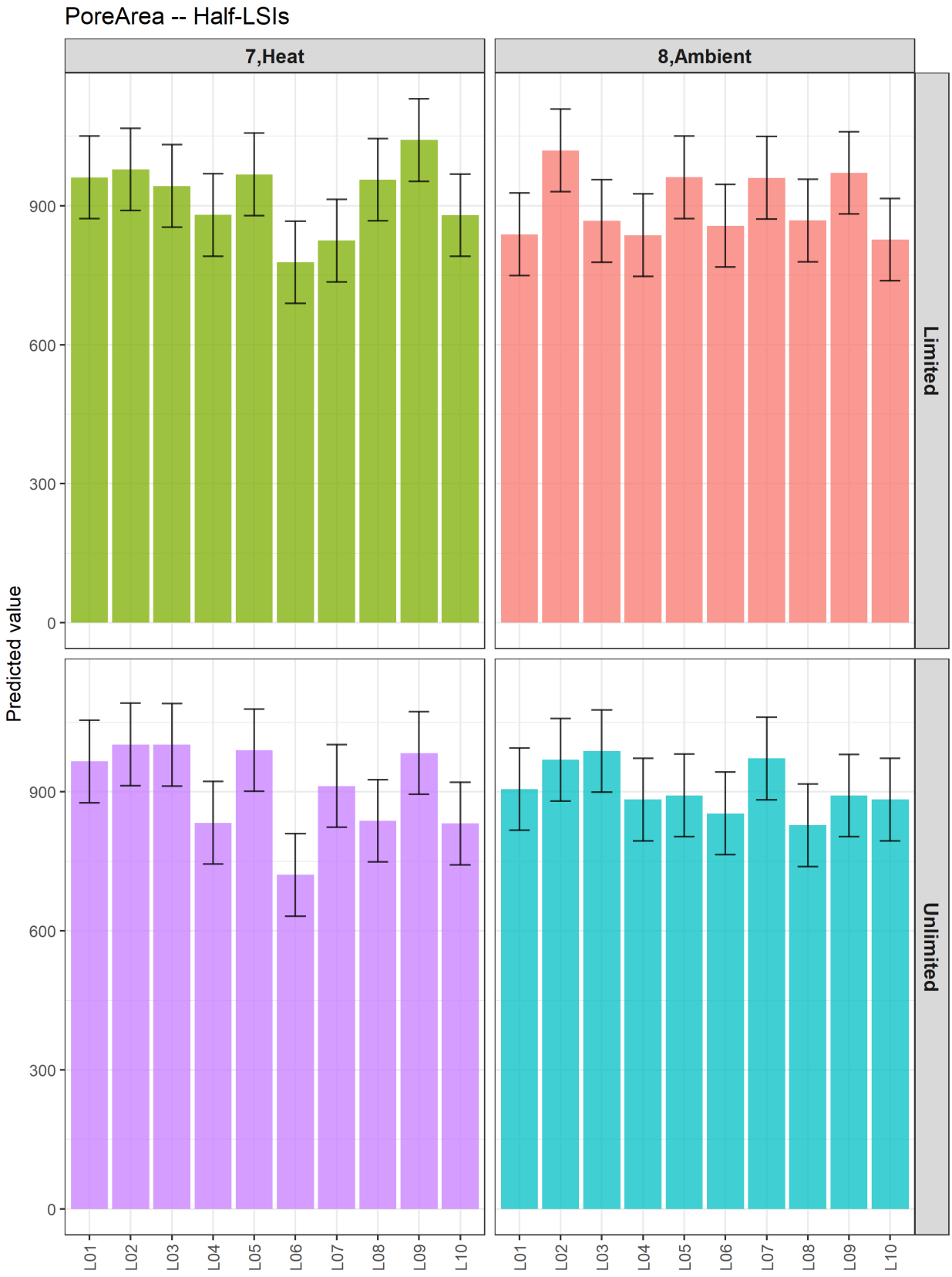


Figure 18: plot of chunk show-v2

```
     HLine)
random: ~ Room:Side:Location
residual: NULL
```

```
####  Summary of the fitted variance parameters for gsmax
```

```
                      component   std.error   z.ratio bound %ch
Room:Side:Location 0.0008122171 0.002065471 0.3932358     P   0
units!R            0.0336374469 0.004475327 7.5161981     P   0
```

```
####  Pseudo-anova table for fixed terms


Wald tests for fixed effects.
Response: gsmax
```

	Df	denDF	F.inc	F.con	Margin	Pr
(Intercept)	1	3	5752.0	5752.0		0.0000
ASide	1	3	3.1	3.0	B	0.1826
ASide:ABlock	2	3	3.8	3.9	C	0.1468
Rm_Temperature	1	3	0.4	0.4	A	0.5767
Rm_Temperature:AWatering	1	113	1.3	1.3	B	0.2581
Rm_Temperature:ALine	9	113	1.1	1.1	B	0.3636
Rm_Temperature:AWatering:ALine	9	113	1.8	1.8	C	0.0777
Rm_Temperature:HWatering	1	113	1.4	1.6	B	0.2153
Rm_Temperature:HLine	9	113	2.5	2.5	B	0.0106
Rm_Temperature:HWatering:HLine	9	113	0.7	0.7	C	0.7318

```
####  Sequence of model investigations for gsmax

(If a row has NA for p but not denDF, DF and denDF relate to fixed and variance parameter numbers)
```

	terms	DF	denDF	p
1	Initial model	43	2	NA
2	Start with residual heterogeneity in Watering & Rm_Temperature for Residual	NA	NA	NA
3	Try homogeneity in Rm_Temperature for Residual	0	-2	NA
4	Try homogeneity in Watering & Rm_Temperature for Residual	0	-1	NA
5	Room:Side	1	NA	NA
6	Room:Side:Location:SubUnit	1	NA	NA
7	Fix bound variance components	0	0	NA
8	Rm_Temperature:HWatering:HLine	9	113	0.7318
9	Rm_Temperature:HLine	9	113	0.0106
10	Rm_Temperature:HWatering	1	113	0.2153
11	Rm_Temperature:AWatering:ALine	9	113	0.0777
12	Rm_Temperature:ALine	9	113	0.3636
13	Rm_Temperature:AWatering	1	113	0.2581

	AIC	BIC	action
1	-330.210392	-192.109703	Starting model
2	NA	NA	Changed residual

```
3    -1.238629   -6.745809          Swapped
4    -1.724391   -4.477981          Swapped
5           NA          NA         Boundary
6           NA          NA         Boundary
7     0.000000    0.000000          Swapped
8           NA          NA   Nonsignificant
9           NA          NA      Significant
10          NA          NA   Nonsignificant
11          NA          NA   Nonsignificant
12          NA          NA   Nonsignificant
13          NA          NA   Nonsignificant

  Information criteria for spatial models

NULL

Chosen terms
[1] "Rm_Temperature:HLine"

LSDs
          c    minLSD   meanLSD  maxLSD assignedLSD accuracyLSD falsePos falseNeg
overall 780 0.2568615 0.2595447 0.28097   0.2595447  0.08254961        1        3

Joining with `by = join_by(Rm_Temperature, Watering, Line.ID)`
Joining with `by = join_by(Watering, Line.ID, Rm_Temperature)`
```

MODEL SUMMARIES

Wald tables

```
for (resp in responses.anat[kvec]) {
  tbl <- models[[resp]]$wald.tab[-1,]
  cat("\n\n", resp)
  print(kable(tbl, digits = 3))
}
```

```
 StomatalCount

|                               | Df| denDF| F.inc| F.con|Margin |    Pr|
|:------------------------------|--:|-----:|-----:|-----:|:------|-----:|
|ASide                          |  1| 101.6| 2.202| 2.017|B      | 0.159|
|ASide:ABlock                   |  2| 101.7| 5.658| 5.984|C      | 0.003|
|Rm_Temperature                 |  1| 104.0| 0.079| 0.079|A      | 0.779|
|Rm_Temperature:AWatering       |  1|  58.9| 0.942| 0.942|B      | 0.336|
|Rm_Temperature:ALine           |  9|  58.9| 0.872| 0.872|B      | 0.555|
|Rm_Temperature:AWatering:ALine |  9|  58.9| 0.827| 0.827|C      | 0.594|
|Rm_Temperature:HWatering       |  1|  56.8| 0.910| 1.064|B      | 0.307|
|Rm_Temperature:HLine           |  9|  56.8| 7.074| 7.074|B      | 0.000|
|Rm_Temperature:HWatering:HLine |  9|  56.8| 1.692| 1.692|C      | 0.112|


 StomatalDensity
```

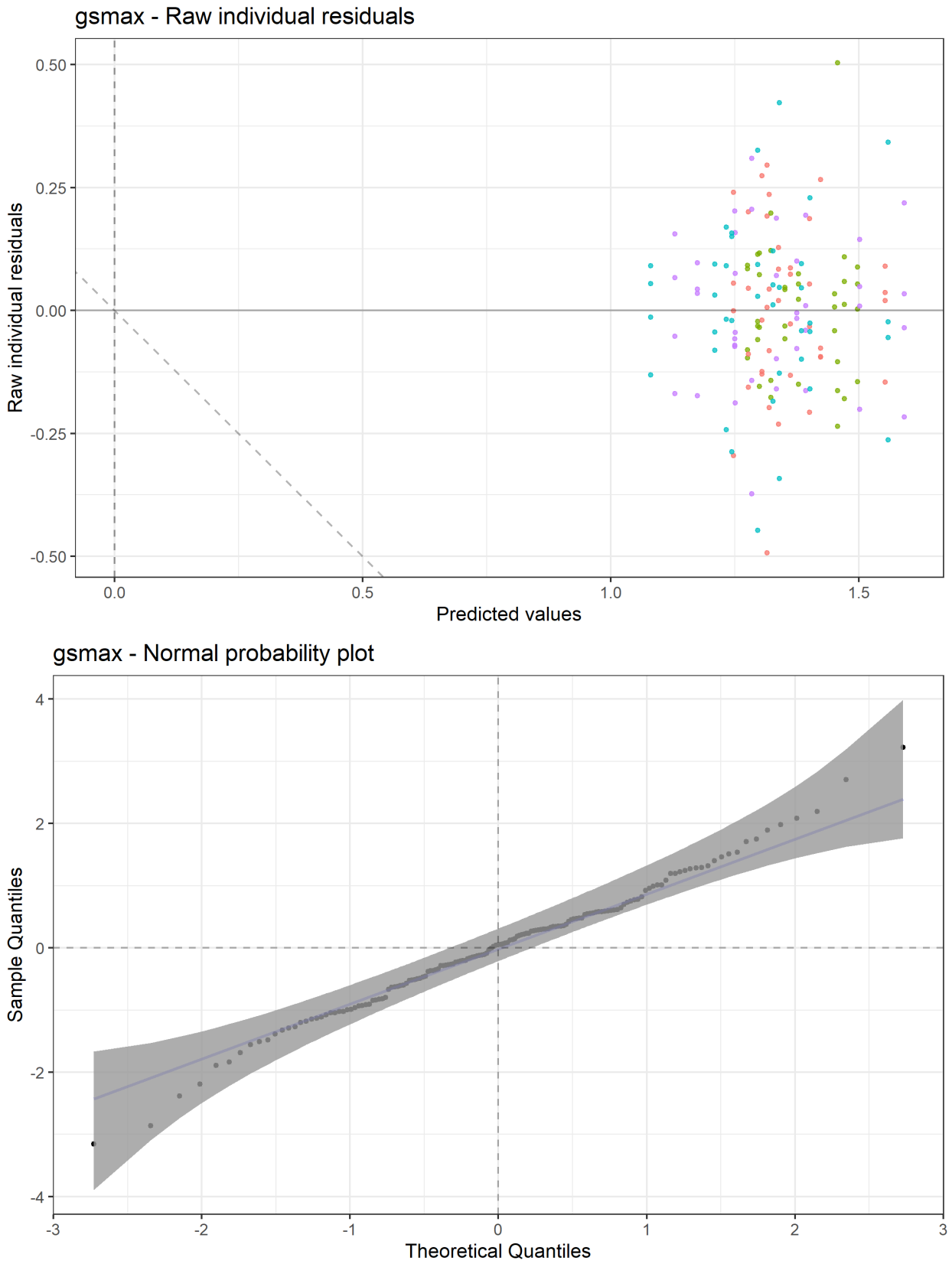


Figure 19: plot of chunk show-v2

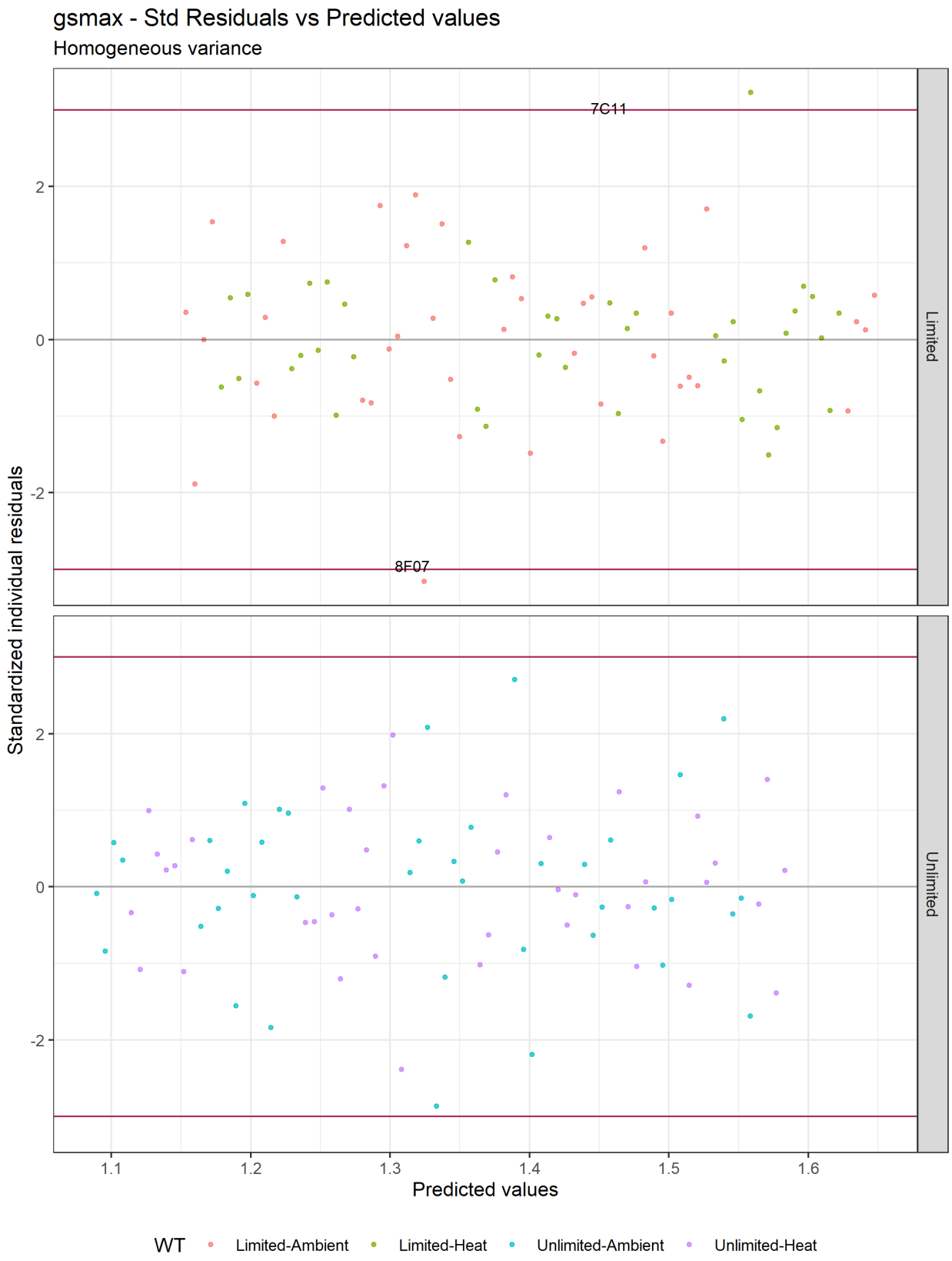


Figure 20: plot of chunk show-v2

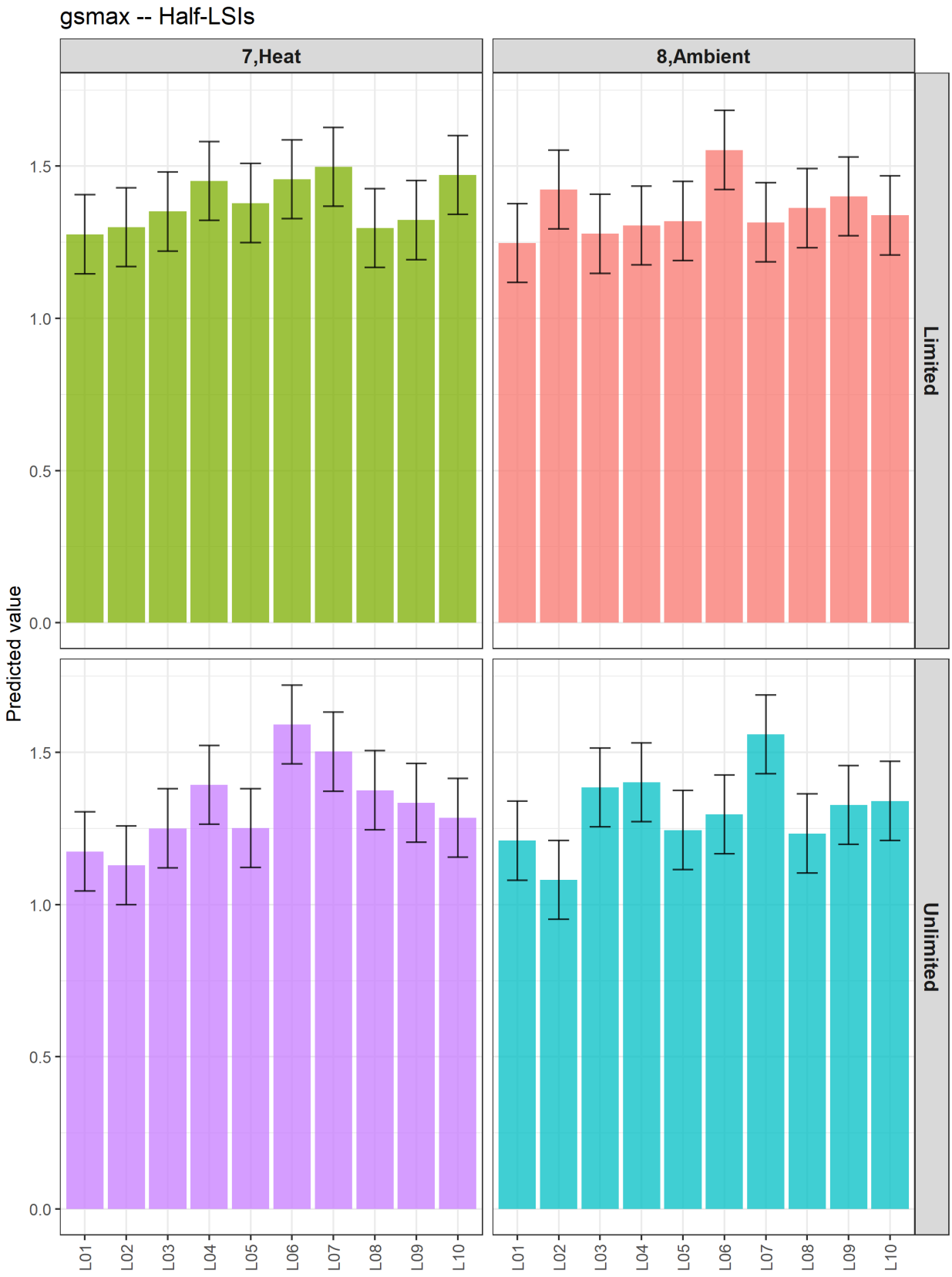


Figure 21: plot of chunk show-v2

	Df	denDF	F.inc	F.con	Margin	Pr
ASide	1	101.6	2.202	2.017	B	0.159
ASide:ABlock	2	101.7	5.658	5.984	C	0.003
Rm_Temperature	1	104.0	0.079	0.079	A	0.779
Rm_Temperature:AWatering	1	58.9	0.942	0.942	B	0.336
Rm_Temperature:ALine	9	58.9	0.872	0.872	B	0.555
Rm_Temperature:AWatering:ALine	9	58.9	0.827	0.827	C	0.594
Rm_Temperature:HWatering	1	56.8	0.910	1.064	B	0.307
Rm_Temperature:HLine	9	56.8	7.074	7.074	B	0.000
Rm_Temperature:HWatering:HLine	9	56.8	1.692	1.692	C	0.112

StomatalArea

	Df	denDF	F.inc	F.con	Margin	Pr
ASide	1	9.0	2.643	2.493	B	0.149
ASide:ABlock	2	9.0	0.027	0.027	C	0.974
Rm_Temperature	1	9.0	0.373	0.373	A	0.556
Rm_Temperature:AWatering	1	8.9	2.588	2.599	B	0.142
Rm_Temperature:ALine	9	106.1	3.437	3.437	B	0.001
Rm_Temperature:AWatering:ALine	9	106.1	1.446	1.445	C	0.178
Rm_Temperature:HWatering	1	9.1	1.410	1.280	B	0.287
Rm_Temperature:HLine	9	107.6	3.360	3.360	B	0.001
Rm_Temperature:HWatering:HLine	9	107.6	0.247	0.247	C	0.986

StomatalWidth

	Df	denDF	F.inc	F.con	Margin	Pr
ASide	1	1.0	2.416	2.226	B	0.376
ASide:ABlock	2	8.0	0.075	0.106	C	0.901
Rm_Temperature	1	1.0	0.261	0.261	A	0.699
Rm_Temperature:AWatering	1	7.9	7.614	7.648	B	0.025
Rm_Temperature:ALine	9	106.7	2.419	2.419	B	0.015
Rm_Temperature:AWatering:ALine	9	106.8	2.294	2.294	C	0.021
Rm_Temperature:HWatering	1	8.1	6.725	5.995	B	0.040
Rm_Temperature:HLine	9	106.8	3.090	3.090	B	0.002
Rm_Temperature:HWatering:HLine	9	106.8	0.234	0.234	C	0.989

StomatalLength

	Df	denDF	F.inc	F.con	Margin	Pr
ASide	1	9.0	0.348	0.354	B	0.567
ASide:ABlock	2	9.0	0.569	0.576	C	0.581
Rm_Temperature	1	9.0	0.209	0.209	A	0.659
Rm_Temperature:AWatering	1	8.9	0.047	0.046	B	0.834
Rm_Temperature:ALine	9	107.0	1.481	1.481	B	0.164
Rm_Temperature:AWatering:ALine	9	107.0	0.494	0.494	C	0.876
Rm_Temperature:HWatering	1	9.1	0.136	0.102	B	0.757

|Rm_Temperature:HLine | 9| 106.9| 3.554| 3.554|B | 0.001|
|Rm_Temperature:HWatering:HLine | 9| 107.0| 0.448| 0.448|C | 0.906|

 PoreArea

	Df	denDF	F.inc	F.con	Margin	Pr
ASide	1	3.0	0.391	0.400	B	0.572
ASide:ABlock	2	3.0	0.511	0.519	C	0.640
Rm_Temperature	1	3.0	0.181	0.181	A	0.699
Rm_Temperature:AWatering	1	6.0	0.033	0.032	B	0.864
Rm_Temperature:ALine	9	106.9	1.406	1.406	B	0.195
Rm_Temperature:AWatering:ALine	9	106.9	0.560	0.560	C	0.827
Rm_Temperature:HWatering	1	6.1	0.163	0.123	B	0.738
Rm_Temperature:HLine	9	106.9	3.520	3.520	B	0.001
Rm_Temperature:HWatering:HLine	9	106.9	0.509	0.509	C	0.865

 gsmax

	Df	denDF	F.inc	F.con	Margin	Pr
ASide	1	3	3.103	2.983	B	0.183
ASide:ABlock	2	3	3.770	3.891	C	0.147
Rm_Temperature	1	3	0.390	0.390	A	0.577
Rm_Temperature:AWatering	1	113	1.292	1.292	B	0.258
Rm_Temperature:ALine	9	113	1.107	1.107	B	0.364
Rm_Temperature:AWatering:ALine	9	113	1.790	1.790	C	0.078
Rm_Temperature:HWatering	1	113	1.450	1.553	B	0.215
Rm_Temperature:HLine	9	113	2.546	2.546	B	0.011
Rm_Temperature:HWatering:HLine	9	113	0.673	0.673	C	0.732

Chosen terms

```
tableChosenModels(models, responses.anat)
```

```
  Trait
1 StomatalCount
2 StomatalDensity
3 StomatalArea
4 StomatalWidth
5 StomatalLength
6 PoreArea
7 gsmax
  Chosen.model
1 Rm_Temperature:HLine
2 Rm_Temperature:HLine
3 Rm_Temperature:HLine + Rm_Temperature:ALine
4 Rm_Temperature:HLine + Rm_Temperature:HWatering + Rm_Temperature:AWatering:ALine
5 Rm_Temperature:HLine
6 Rm_Temperature:HLine
7 Rm_Temperature:HLine
```

Table the residual variances

```
tableVarModels(models, responses.anat)
```

```
  Trait           Variance.model
1 StomatalCount   Variance differs by Rm_Temperature
2 StomatalDensity Variance differs by Rm_Temperature
3 StomatalArea    Variance differs by Rm_Temperature
4 StomatalWidth   Variance differs by Watering
5 StomatalLength  Variance differs by Watering
6 PoreArea        Variance differs by Watering
7 gsmax           Homogeneous variance
```

Extremal residuals

```
typelabls <- c("Anat")
resptlabs <- rep(typelabls,
                 times = length(responses.anat))
print(cbind(responses.anat, resptlabs), quote=F)
```

```
     responses.anat  resptlabs
[1,] StomatalCount   Anat
[2,] StomatalDensity Anat
[3,] StomatalArea    Anat
[4,] StomatalWidth   Anat
[5,] StomatalLength  Anat
[6,] PoreArea        Anat
[7,] gsmax           Anat
```

```
res.lim.label <- 5
Outliers.dat <- data.frame()
tbl <- data.frame()
for (resp in responses.anat[kvec])
{
  if(resp %in% responses.pot){
    tmp.dat <- models[[resp]]$asreml.obj$call$data
    res.dat <- within(tmp.dat, res.std <- residuals(models[[resp]]$asreml.obj, type="stdCond"))
    res.dat <- res.dat[!is.na(res.dat$res.std), ]
    outliers.dat <- subset(res.dat, abs(res.std) > res.lim.label)
    outliers.dat <- outliers.dat[c("Cell.ID", "res.std", "Rm_Temperature",
                                   "Watering", "Line")]
    if (nrow(outliers.dat) > 0)
    {
      Outliers.dat <- rbind(Outliers.dat,
                            data.frame(Response=resp, outliers.dat))
      tbl <- rbind(tbl,
                   data.frame(Response = resp,
                              n        = nrow(outliers.dat),
                              rmin     = min(res.dat$res.std, na.rm=T),
                              rmax     = max(res.dat$res.std, na.rm=T)))
    }
  } else{
    tmp.dat <- models[[resp]]$asreml.obj$call$data
    res.dat <- within(tmp.dat, res.std <- residuals(models[[resp]]$asreml.obj, type="stdCond"))
```

```
    res.dat <- res.dat[!is.na(res.dat$res.std), ]
    outliers.dat <- subset(res.dat, abs(res.std) > res.lim.label)
    outliers.dat <- outliers.dat[c("Cell.ID", "Stomata", "res.std", "Rm_Temperature",
                                   "Watering", "Line")]
    if (nrow(outliers.dat) > 0)
    {
      Outliers.dat <- rbind(Outliers.dat,
                            data.frame(Response=resp, outliers.dat))
      tbl <- rbind(tbl,
                   data.frame(Response = resp,
                              n        = nrow(outliers.dat),
                              rmin     = min(res.dat$res.std, na.rm=T),
                              rmax     = max(res.dat$res.std, na.rm=T)))
    }
  }
}

if (nrow(Outliers.dat)==0) {

  cat("\n\n #### No outliers found\n\n")

} else {

  Outliers.dat <- with(Outliers.dat, Outliers.dat[order(Cell.ID, Stomata, Response),])
  Outliers.dat <- within(Outliers.dat, res <- round(res.std,2))
  tmp.dat <- Outliers.dat[c("Cell.ID","Stomata", "res","Response", "Rm_Temperature",
                            "Watering", "Line")]
  names(tmp.dat)[1] <- c("Plant")
  outliers.dat <- unique(tmp.dat[,-(3:4)])
  tbl <- within(tbl, {
    rmin <- ifelse(rmin < 0, round(rmin,2), NA)
    rmax <- ifelse(rmax > 0, round(rmax,2), NA)
  })

  msg1 <- paste("Standardised residuals outside the range (-",
                res.lim.label, ", ",res.lim.label,")",
                sep="")
  msg2 <- "Unique plants in outlier list"

  cat("\n\n #### Number of outliers per response\n\n"); print(tbl, right=F)
  cat("\n\n",msg1,"\n\n"); print(tmp.dat, right=F)
  cat("\n\n",msg2,"\n\n"); print(outliers.dat, right=F)

  tmp2.dat <- within(tmp.dat, {
    Magnitude <- abs(res)
    Direction <- factor(sign(res), levels = c(1,-1), labels = c("pos","neg"))
    Response <- factor(Response, levels = responses.anat)
    Resptype <- dae::fac.recode(Response, newlevels = resptlabs, levels = rev(typelabls))
  })

  lims <- c(res.lim.label,
            max(4, 0.5*ceiling(max(tmp2.dat$Magnitude)/0.5)))  #round to nearest half
  plt <- ggplot(tmp2.dat, aes(Plant, as.character(Response),
```

```
                              shape=Direction, colour=Magnitude, fill=Magnitude)) +
    facet_grid(rows = vars(Resptype), space="free", scales="free") +
    geom_point(size=2) +
    scale_colour_gradient(low="green", high="red", limits = lims) +
    scale_fill_gradient(low="green", high="red", limits = lims) +
    scale_shape_manual(values = c(24,25), drop=F) +
    labs(y=NULL) +
    scale_y_discrete(drop=F) +
    ggtitle("Outliers by Plant and Response",
            subtitle = "Coloured by magnitude of standardized residual") +
    theme_bw() +
    theme(axis.text.x = element_text(angle=90, hjust=1, vjust=0.5),
          strip.text = element_text(size=10, face="bold"))

  print(plt)
}
```

Number of outliers per response

	Response	n	rmin	rmax
1	StomatalArea	2	-4.84	5.49
2	StomatalWidth	2	-4.83	5.06
3	StomatalLength	2	-3.50	5.99
4	PoreArea	3	-3.07	7.83

Standardised residuals outside the range (-5, 5)

	Plant	Stomata	res	Response	Rm_Temperature	Watering	Line
26662	CER7A03	28	5.74	PoreArea	7,Heat	Limited	PBI16C009-0C-0N-0N-010N-14N
2666	CER7A03	28	5.29	StomatalArea	7,Heat	Limited	PBI16C009-0C-0N-0N-010N-14N
26661	CER7A03	28	5.60	StomatalLength	7,Heat	Limited	PBI16C009-0C-0N-0N-010N-14N
3142	CER7A07	29	5.03	StomatalWidth	7,Heat	Limited	PBI19N007-0N-3N
412	CER7D09	31	5.06	StomatalWidth	7,Heat	Unlimited	PBI17N002-0C-0N-0N-010N-16N
3315	CER7E10	23	5.28	PoreArea	7,Heat	Unlimited	PBI19N007-0N-3N
36581	CER7F07	16	7.83	PoreArea	7,Heat	Limited	PBI17N007-0C-0N-0N-010N-8N
3658	CER7F07	16	5.99	StomatalLength	7,Heat	Limited	PBI17N007-0C-0N-0N-010N-8N
2009	CER7F14	26	5.49	StomatalArea	7,Heat	Unlimited	PBI17N013-0C-0N-0N-010N-8N

Unique plants in outlier list

	Plant	Stomata	Rm_Temperature	Watering	Line
26662	CER7A03	28	7,Heat	Limited	PBI16C009-0C-0N-0N-010N-14N
3142	CER7A07	29	7,Heat	Limited	PBI19N007-0N-3N
412	CER7D09	31	7,Heat	Unlimited	PBI17N002-0C-0N-0N-010N-16N
3315	CER7E10	23	7,Heat	Unlimited	PBI19N007-0N-3N
36581	CER7F07	16	7,Heat	Limited	PBI17N007-0C-0N-0N-010N-8N
2009	CER7F14	26	7,Heat	Unlimited	PBI17N013-0C-0N-0N-010N-8N

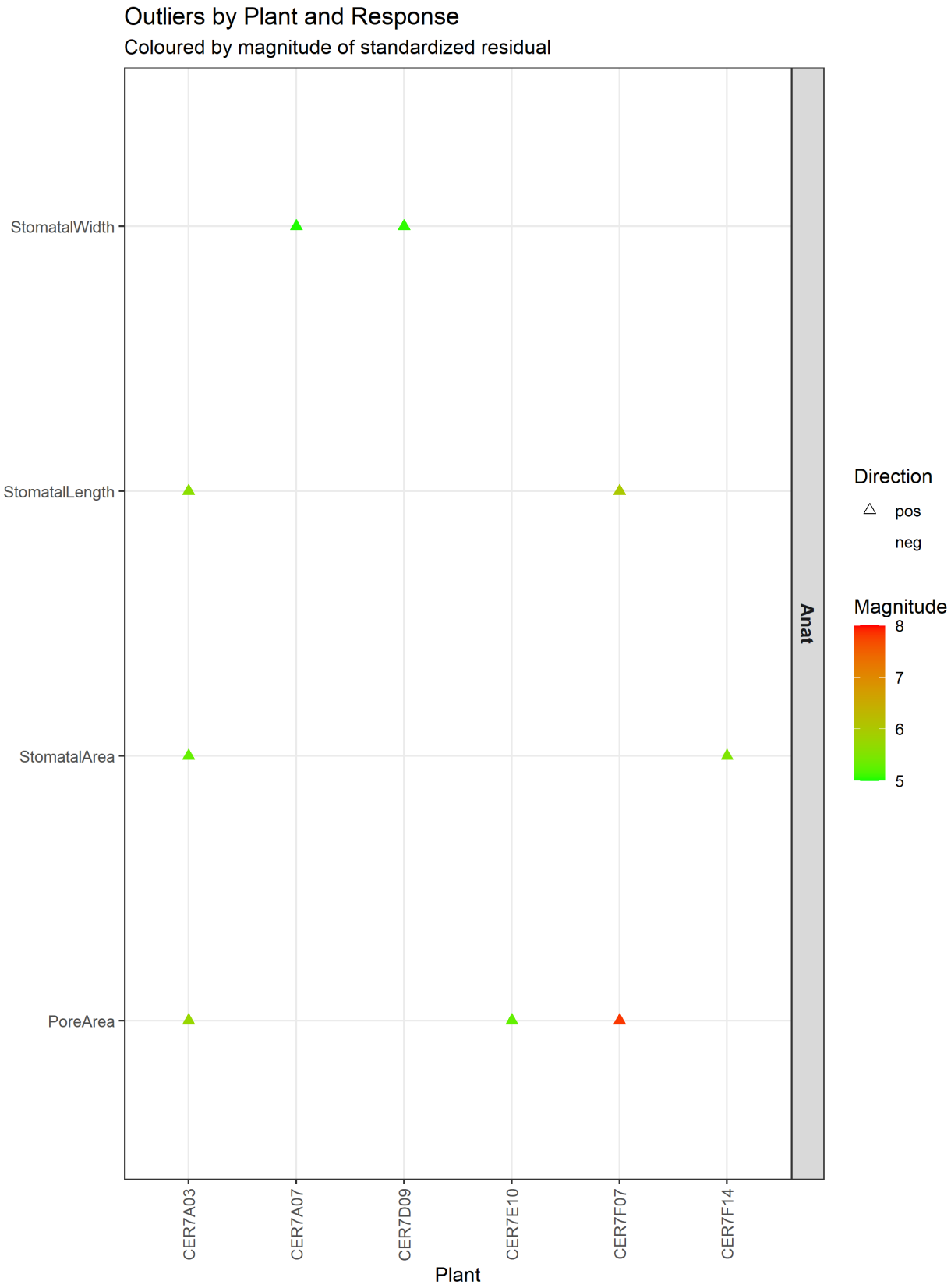


Figure 22: plot of chunk outliers-v2